ONE-RING BOURGAIN ALTERNATIVE AND A 5-DECIMAL BOUND ON THE DIMENSION OF HARMONIC MEASURE

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Abstract. Improving theorems of Bourgain and Badger–Genschaw, we prove that the Hausdorff dimension of harmonic measure on any domain in $\mathbb{R}^3$, $\mathbb{R}^4$, and $\mathbb{R}^n$ for $n \geq 3$ is at most $3 - 10^{-5}$, $4 - 10^{-7}$, and $n - (n-1)^{-(2n+6.12\sqrt{n})}$, respectively. The previous bounds in dimensions 3 and 4 were $3 - 10^{-15}$ and $4 - 10^{-26}$, while Bishop conjectures that the sharp bound is $n - (n-1)^{-1}$. The main difficulty in optimizing Bourgain's method is estimating non-additive s-dimensional Hausdorff-type contents when $s < n$ is close to n. Previous arguments used net contents associated to a fixed m-adic lattice, with m large depending on n. We redevelop the dimension theory using pairwise disjoint cubes in general position and obtain a new estimate showing that one thin ring without large holes surrounding a cube makes it unlikely for Brownian motion to reach the cube before exiting the domain. A key feature of the construction is that the holes are round rather than rectangular. We also propose ten challenges on the dimension of harmonic measure that may provide accessible targets en route to Bishop's conjecture.

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1. Introduction

1.1. **Background.** One way to understand harmonic measure is as the exit distribution of Brownian motion from a domain. Let $\Omega \subset \mathbb{R}^n$ be a bounded, connected, open set and let $x \in \Omega$. Standard Brownian motion $B : [0, \infty) \to \mathbb{R}^n$ started at $B(0) = x$ eventually

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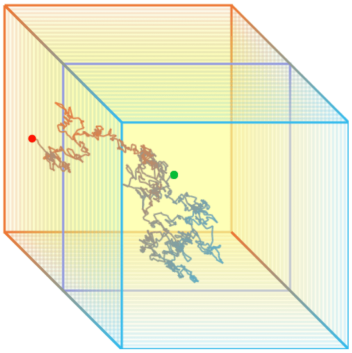


FIGURE 1.1. Brownian motion simulated in a cube, starting at the green dot and exiting at the red dot. With its pole at the center, each face of the cube has harmonic measure 1/6, while each edge has harmonic measure 0.

exits Ω at a random point $B(t_{\text{exit}}) \in \partial\Omega$. The probability that the exit point belongs to a set $E \subset \partial\Omega$ is called the *harmonic measure* of E in Ω with *pole* at x and is denoted by $\omega_\Omega^x(E)$. Thus, with Ω and x fixed, $E \mapsto \omega_\Omega^x(E)$ is a Borel probability measure on $\partial\Omega$. Kakutani [Kak44] proved the complementary analytic fact that, with Ω and E fixed, the function $x \mapsto \omega_\Omega^x(E)$ is harmonic in Ω, and either $\omega_\Omega^x(E) = 0$ for every $x \in \Omega$ or $\sup_{x\in\Omega} \omega_\Omega^x(E) = 1$. In particular, the null sets of ω_Ω^x do not depend on the pole, and we commonly write that a property holds ω_Ω-a.e. or ω-a.e. without specifying x.

Harmonic measure also represents solutions of the Dirichlet problem. For every real-valued $f \in C(\partial\Omega)$, the function

$$u(x) := \int_{\partial\Omega} f(y)\, d\omega_\Omega^x(y), \quad \text{for all } x \in \Omega, \tag{1.1}$$

is harmonic in Ω and satisfies

$$\lim_{x\to y} u(x) = f(y) \quad \text{at } \omega_\Omega\text{-a.e. } y \in \partial\Omega. \tag{1.2}$$

More precisely, (1.2) holds outside a set of logarithmic capacity zero when $n = 2$ and Newtonian capacity zero when $n \geq 3$, hence outside a set of Hausdorff dimension at most $n - 2$. Equations (1.1) and (1.2), together with the probabilistic interpretation above, provide most of the background on harmonic measure needed in this paper. For more extensive introductions, see Bishop's review [Bis07] of the book of Garnett and Marshall [GM05], as well as Kenig [Ken94], Heinonen *et al.* [HKM06], Helms [Hel09], Mörters and Peres [MP10], and the forthcoming book of Prats and Tolsa [PT26].

We are interested in the geometric size of harmonic measure. Recall that the *upper Hausdorff dimension* of a measure is the least Hausdorff dimension of a set of full measure, while the *lower Hausdorff dimension* is the infimum of the Hausdorff dimensions of sets of positive measure. The Hausdorff dimensions of $\omega_\Omega = \omega_\Omega^x$ depend on the domain but not on the choice of pole. On any bounded domain in $\mathbb{R}^n$, the lower Hausdorff dimension of harmonic measure is at least $n - 2$, see e.g. [Bad11, Appendix A], [Wu86]. On a smooth or Lipschitz domain, harmonic measure is mutually absolutely continuous with respect to surface measure on $\partial\Omega$ and thus has Hausdorff dimension equal to $n - 1$ [Dah77]; see Figure 1.1. By contrast, on general domains, whose boundaries can be rough and have Hausdorff dimension greater than $n-1$, the dimension of harmonic measure may be either

larger or smaller than $n-1$, yet it can never reach the ambient dimension n. The modern higher-dimensional theory begins with the following two results.

Theorem A (Wolff [Wol95][1]). *For all $n \geq 3$, there exist simply connected NTA[2] domains $\Omega \subset \mathbb{R}^n$ such that the lower and upper Hausdorff dimensions of harmonic measure agree and are $> n-1$. There also exist simply connected NTA domains $\Omega \subset \mathbb{R}^n$ such that the lower and upper Hausdorff dimensions of harmonic measure agree and are $< n-1$.*

Theorem B (Bourgain [Bou87]). *For all $n \geq 3$, there exists $b_n > 0$ such that the upper Hausdorff dimension of harmonic measure on any domain in $\mathbb{R}^n$ is at most $n - b_n$.*

Theorem B implies that whenever $\dim_H \partial\Omega > n - b_n$, harmonic measure is carried by a set $E \subset \partial\Omega$ with $\dim_H E < \dim_H \partial\Omega$. In probabilistic terms, Brownian motion almost surely exits Ω through a set that is geometrically smaller than the full boundary. This phenomenon, called *dimension drop*, was observed by Kaufman and Wu [KW85] for the domain bounded by a von Koch snowflake and by Carleson [Car85] for Cantor-type planar domains, and its study has grown substantially in recent years. For classical harmonic measure, see [Vol93], [Bat96], [BZ15], [Azz20], [DJJ23], [Tol24], [CT26]; for p-harmonic measures with $p \neq 2$, see [LNV13], [ALV15]; for elliptic measures, see [Swe92], [Swe94], [DM21], [Per25], [GMPT25], [GM25]; for Robin harmonic measures, see [DDE$^+$24]; and for caloric and parabolic measures, see [Swe96], [BG25]. For further results related to Theorem A, see [LVV05], [KPT09], [Bad12], [Azz18].

Let d_n denote the supremum of the upper Hausdorff dimensions of harmonic measures over all domains in $\mathbb{R}^n$. The optimal constant in Theorem B,

$$b_n := n - d_n, \tag{1.3}$$

is called *Bourgain's constant* for harmonic measure in $\mathbb{R}^n$. Thus, the theorems of Bourgain and Wolff imply $0 < b_n < 1$ for every $n \geq 3$. Moreover, if $\Omega \subset \mathbb{R}^n$, then the cylinder $\Omega \times \mathbb{R} \subset \mathbb{R}^{n+1}$ satisfies $\overline{\dim}_H \omega_{\Omega\times\mathbb{R}} = \overline{\dim}_H \omega_\Omega + 1$. Consequently,

$$d_n + 1 \leq d_{n+1} \qquad \text{and} \qquad b_n \geq b_{n+1} \quad \text{for all } n \geq 2. \tag{1.4}$$

The planar theory provides an important baseline. Using conformal mappings and subharmonicity of $\log|f|$ for holomorphic functions f, Jones and Wolff [JW88] proved that $b_2 = 1$. Makarov [Mak85] proved that on every simply connected planar domain the lower and upper Hausdorff dimensions of harmonic measure are exactly 1. These planar results helped motivate the theorems of Bourgain and Wolff in higher dimensions.

For $n \geq 3$, the exact value of b_n is unknown. The case $n = 3$ is especially compelling, both as the first unknown dimension and because of its connection with diffusion in physical space [FGAS08]. Bishop [Bis92] conjectures that $b_3 = 1/2$, and more generally that $b_n = 1/(n-1)$ for every $n \geq 3$. More than thirty years later, even the order of

[1]Although published in 1995, preliminary versions of Wolff's paper circulated in the 1980s.

[2]The non-tangentially accessible domains of Jerison and Kenig are quantitatively open and quantitatively connected; they include Lipschitz domains and many fractal domains [JK82], [KT97].

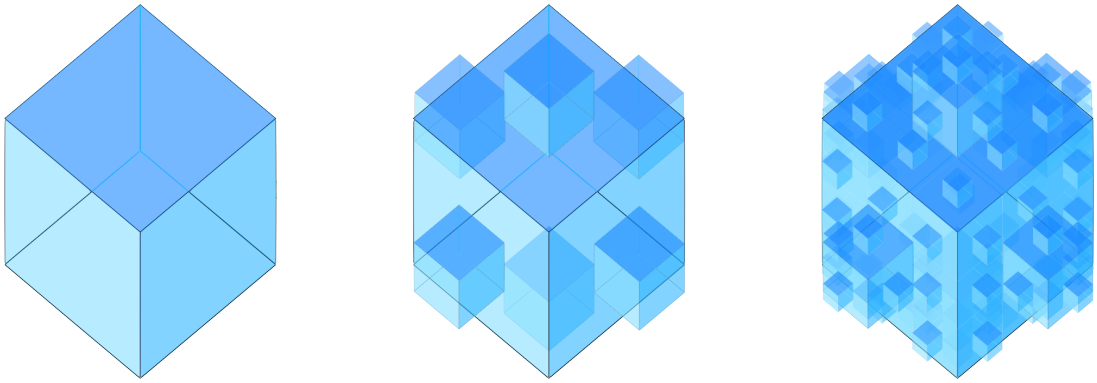


FIGURE 1.2. Levels 0, 1, and 2 of the cubical von Koch snowball domain $G \subset \mathbb{R}^3$ examined in [GLFS05]. Each square of side length 3^{-l} in the level l boundary carries an outward cube of side length $3^{-(l+1)}$, and G is the interior of the union of the finite-level approximations. Here $\dim_H \partial G = \log_3 13$, while the numerics of Grebenkov *et al.* give $\dim_H \omega_G = 2.007 \pm 0.002$.

magnitude of b_n as $n \to \infty$ remains unknown. A numerical experiment of Grebenkov *et al.* [GLFS05] on the cubical von Koch snowball in Figure 1.2 suggests that $b_3 \leq 0.995$; the first challenge in §2 calls for independently reproducing or refuting this estimate and for numerical investigations of harmonic measure on other explicit domains in $\mathbb{R}^3$. More recently, Badger and Genschaw revisited Bourgain's proof and obtained the first explicit lower bounds on b_n in dimensions $n \geq 3$.

Let log denote the natural logarithm.

Theorem C (Badger–Genschaw [BG24])**.** *We have* $b_3 \geq 1 \times 10^{-15}$, $b_4 \geq 2 \times 10^{-26}$, *and* $b_n \geq 0.6n^{-2n(n-1)}/\log n$ *for all* $n \gg 3$. *In particular, the upper Hausdorff dimension of harmonic measure on any domain in* $\mathbb{R}^3$ *is at most* $2.99999\,99999\,99999$.

For comparison with Bishop's conjecture and the new estimates below, Theorem C implies that there exists some n_0 (uncomputed) such that

$$b_n \geq \left(\frac{1}{n-1}\right)^{2n^2} \quad \text{for all } n \geq n_0.$$

1.2. **New results and proof strategy.** The main result of this paper improves both the explicit low-dimensional bounds in Theorem C and the asymptotic exponent as $n \to \infty$. We also obtain the first fully effective bound in arbitrary dimensions.

Theorem 1.1. *For all* $n \geq 2$,

$$b_n \geq \left(\frac{1}{n-1}\right)^{2n+6.12\sqrt{n}}. \tag{1.5}$$

Corollary 1.2. *We have* $b_3 \geq 2^{-(6+6.12\sqrt{3})} = 1.0065\ldots \times 10^{-5}$. *Thus, the upper Hausdorff dimension of harmonic measure on any domain in* $\mathbb{R}^3$ *is less than* 2.99999.

The proof follows Bourgain's original outline, which is based on a local dichotomy. Roughly speaking, at each scale either the boundary has a sufficiently large hole, yielding an efficient cover, or the absence of such holes creates a barrier that makes it difficult for Brownian motion to pass deeper into the domain. Previous implementations of the second alternative required many concentric barrier rings. Rather than optimizing a single step, we rework every stage of the argument—replacing coarse estimates by sharp computations, allowing the covering geometry to be chosen adaptively, and introducing new barrier estimates—to eliminate accumulated quantitative losses and arrive at what we believe is the essentially optimal form of Bourgain's alternative.

On the geometric side, Section 3 develops quantitative principles for s-dimensional Hausdorff contents when $s < n$, where non-additivity makes the efficiency of a cover sensitive to the scales of its cubes: a hole in a cubical region can favor a mixed-scale cover when s is close to n, while a single cube may be more efficient as s decreases. A general treatment of such calculations does not appear to exist elsewhere. In §4, we replace the fixed m-adic lattice used previously by a Bourgain-type dimension lemma that permits pairwise disjoint covering cubes in general position and allows the entropy order $\alpha \in (0, 1)$ to be chosen to sharpen the estimate. In §5, we develop a method of *approximate polar integration* that transfers radial integral estimates against Frostman measures to estimates against Lebesgue measure with explicit control of the error. Our method yields a nearly sharp Bourgain estimate of the local harmonic measure in round regions.

The argument culminates in the *one-ring alternative* of §6. At a given scale, choose a thin ring. If the boundary has a sufficiently large hole in the ring, then the hole yields an efficient mixed-scale cover and relative Hausdorff content below 1. If no such hole exists, the local estimate on harmonic measure makes the ring a barrier to Brownian motion. Then a new computation converts this local barrier into quantitative control of harmonic measure crossing a single ring, replacing the previous multi-ring construction. A further refinement accounts for arbitrarily many returns of Brownian motion to the barrier and sharpens the constant. The resulting reduction in geometric cost allows the Hausdorff-content gain in the large-hole case to remain effective for exponents s farther below n, producing the improved bounds on b_n. These gains are cumulative: reaching the five-decimal estimate in dimension 3 depends on the refinements working together. In §7, an explicit family of parameters yields Theorem 1.1 for $n \geq 6$, while separately optimized explicit choices of parameters gives the stated estimate in dimensions 3, 4, and 5.

2. Ten challenges on dimension of harmonic measure

Given the general lack of progress on computing Bourgain's constants b_n, when $n \geq 3$, we propose ten intermediate challenges in order to spur further research on dimension of harmonic measure. The challenges range from concrete problems that invite direct computation to broader, more open-ended theoretical questions. Some ask for explicit examples or quantitative estimates, while others concern qualitative phenomena and the nature of dimension drop. We start with four challenges in $\mathbb{R}^3$.

Challenge 1 (Numerical Experiment Challenge)**.** Using either the same or different methodology as in [GLFS05], perform an experiment in order to numerically estimate the dimension of harmonic measure on the cubical von Koch snowball domain in Figure 1.2. Estimate the dimension of harmonic measure on other domains in $\mathbb{R}^3$ such as domains whose boundaries are cubical Cantor sets.

Challenge 2 (Explicit Upper Bound Challenge)**.** Prove that $b_3 \leq w < 1$ for some explicit number w. If possible, rigorously verify the numerical bound discussed above.

If $b_3 = 0.5$, as conjectured, then it should be possible to approximate this value from either side. However, the current bounds $10^{-5} \leq b_3 \overset{?}{\leq} 0.995$ are frustratingly far from 0.5. In the authors' view, a resolution of the following challenge would represent a significant accomplishment on par with determining the value of b_3.

Challenge 3 (One Decimal Challenge)**.** For trivial reasons, $b_3 \geq 0.1$ or $b_3 \leq 0.9$, but we don't currently know the truth of either bound. Prove one (or both).

Note that the previous challenges could each be resolved by estimates on the dimension of harmonic measure alone, without ever computing the dimension of an actual domain. A resolution of the following distinct challenge would be very welcome.

Challenge 4 (Exact Dimension Challenge)**.** Construct a domain $\Omega \subset \mathbb{R}^3$ such that $\overline{\dim}_H \omega_\Omega > 2$ and compute $\overline{\dim}_H \omega_\Omega$ exactly.

Let us turn to high-dimensional asymptotics. We know that $b_2 = 1 > b_3$ and $b_n \geq b_{n+1}$ for all $n \geq 3$; see (1.4). We do not currently know whether or not $b_n > b_{n+1}$ for all $n \geq 3$. To aid discussion, for each $n \geq 2$, define the *Wolff constant for harmonic measure* by

$$w_n := \sup_{\Omega \subset \mathbb{R}^n} w_\Omega, \qquad w_\Omega := \overline{\dim}_H \omega_\Omega - (n-1). \tag{2.1}$$

Of course, $w_n + b_n = 1$, but the two constants have different interpretations: w_n measures how much larger the dimension of harmonic measure can be than the classical value $n-1$, while b_n measures how much smaller it must be than the ambient dimension n. A folklore problem is to prove that $\lim_{n\to\infty} b_n = 0$, or equivalently, $\lim_{n\to\infty} w_n = 1$, but remains intractable. The following challenge may serve as a more accessible first step.

Challenge 5 (Growing Excess Challenge)**.** Construct a family of domains $\Omega^n \subset \mathbb{R}^n$, for $n \geq 3$, arising from a single geometric construction, such that

$$0 < w_{\Omega^3} < w_{\Omega^4} < w_{\Omega^5} < \cdots . \tag{2.2}$$

What constitutes a solution to Challenge 5 is open to interpretation. A solution should exclude trivial increases obtained by taking products with Euclidean factors. If you start with domains $\Omega_1, \Omega_2 \subset \mathbb{R}^3$ that satisfy $2 < \overline{\dim}_H \omega_{\Omega_1} < \overline{\dim}_H \omega_{\Omega_2}$ (e.g. built using Wolff's construction), then $\Omega_2 \times \mathbb{R} \subset \mathbb{R}^4$ has $w_{\Omega_2 \times \mathbb{R}} = w_{\Omega_2} > w_{\Omega_1}$, but the pair $\Omega^3 := \Omega_1$ and $\Omega^4 := \Omega_2 \times \mathbb{R}$ does not solve the challenge. The goal is to find a procedure that takes a given domain in $\mathbb{R}^n$ and produces a domain in $\mathbb{R}^{n+1}$ that has larger Wolff excess. Challenge

5 is difficult, because it is not clear which domains should be used. The obvious extension of the cubical von Koch snowball to higher dimensions does not solve the challenge. To wit, if $G^n \subset \mathbb{R}^n$ is obtained from $[0,1]^n$ by repeatedly attaching smaller cubes to the middle-third of each face, then $\overline{\dim}_H \omega_{G^n} \leq \dim_H \partial G^n = \log_3(3^{n-1} + 2n - 1)$. Since the latter numbers are asymptotic to $n-1$, we have $\lim_{n\to\infty} w_{G^n} = 0$. In the authors' opinion, this is also indirect evidence that $\overline{\dim}_H \omega_{G^3}$ is likely to be much smaller than w_3.

An outcome of the present investigation, which culminated in Theorem 1.1, is that an exponential lower bound on Bourgain's constant probably cannot be established with Bourgain's alternative alone. Thus, despite being facially weaker than Bishop's conjecture, a resolution of the following challenge would likely represent significant progress towards understanding the dimension of harmonic measure.

Challenge 6 (Exponential Bound Challenge)**.** Prove $b_n \geq 10^{-Cn}$ for all $n \geq n_0$ for some constants C and n_0, or prove *any* sub-exponential lower bound on b_n when n is large.

Packing measures and packing dimension, introduced by Tricot [Tri82], provide another fundamental notion of size in geometric measure theory. Packing dimension may be viewed as the countable stabilization of a better-known dimension often called upper Minkowski or box-counting dimension; see e.g. [Mat95, Chapter 5]. For each $n \geq 2$, define

$$p_n := n - \sup_{\Omega \subset \mathbb{R}^n} \overline{\dim}_P \omega_\Omega, \tag{2.3}$$

which is the packing-dimension analogue of Bourgain's constant. Since packing dimension dominates Hausdorff dimension, $0 \leq p_n \leq b_n$.

Challenge 7 (Packing Dimension Challenge)**.** Prove or disprove that $p_n > 0$ for every $n \geq 2$. If $p_n > 0$, is $p_n = b_n$? In particular, is $p_2 = 1$?

At present, it is not clear to us whether one should expect a universal packing dimension drop. The evidence is weak and somewhat mixed. On the pro side, Housworth already proved the packing dimension analogue of Makarov's theorem, i.e. harmonic measure on every simply connected planar domain lives on a set of packing dimension 1 [Hou95]. On the other hand, harmonic measure is already known to behave differently with respect to Hausdorff and packing measures. Choi's theorem on the lower 1-density at twist points [Cho04] implies that on many planar domains, including the von Koch snowflake, every set of σ-finite packing measure $\mathcal{P}^1$ has harmonic measure zero. This contrasts with a theorem of Wolff [Wol93] that says for every bounded domain in $\mathbb{R}^2$ there exists a set of σ-finite Hausdorff measure $\mathcal{H}^1$ and full harmonic measure. Additional weaker evidence that p_n might vanish is that the proof of the Jones-Wolff theorem ($b_2 = 1$) relies on a notion of generalized Hausdorff capacity that does not mesh well with packing dimension. Bourgain's method for showing $b_n > 0$ when $n \geq 3$ also cannot be easily modified to give an affirmative answer to Challenge 7, because of its heavy reliance on Frostman's lemma for Hausdorff dimension.

Even if the Bourgain constants were to be computed, there are other natural extremal questions about dimension drop. Define the *maximal dimension drop constant*

$$m_n = \sup_{\Omega \subset \mathbb{R}^n} \left(\dim_H \partial\Omega - \overline{\dim}_H \omega_\Omega \right).$$

There are domains with $\dim_H \partial\Omega = n$, so Bourgain's theorem implies $m_n \geq b_n$.

Challenge 8 (Maximal Dimension Drop Challenge)**.** For each $n \geq 2$, is $m_n = b_n$ or is $m_n > b_n$? If $m_2 > 1$, compute m_2 or obtain nontrivial bounds for it.

A concrete setting for investigating where dimension drop first appears is the class of *Denjoy domains*, i.e. domains $\Omega \subset \mathbb{R}^2$ whose complements are contained in a line. David, Jeznach, and Julia [DJJ23] constructed s-Ahlfors regular Cantor sets $E \subset \mathbb{R}$ with s small for which harmonic measure on $\mathbb{R}^2 \setminus E$ is equivalent to $\mathcal{H}^s|_E$, and hence for which harmonic measure does not have dimension drop. In the opposite direction, Tolsa [Tol24] proved that $\overline{\dim}_H \omega_{\mathbb{R}^2 \setminus E} < s$ whenever $E \subset \mathbb{R}$ is s-Ahlfors regular and $1/2 \leq s < 1$, and asked for the sharp transition between these two phenomena.

Challenge 9 (Denjoy Transition Challenge)**.** Determine the least $s_0 \in (0, 1)$ such that, for every $s \in (s_0, 1)$ and every s-Ahlfors regular compact set $E \subset \mathbb{R}$, the harmonic measure of $\mathbb{R}^2 \setminus E$ satisfies $\overline{\dim}_H \omega_{\mathbb{R}^2 \setminus E} < s$. More generally, determine whether there is an analogous transition for s-Ahlfors regular compact sets $E \subset \mathbb{R}^2$ that are not confined to a line.

We end our list with a challenge concerning elliptic measure, the analogue of harmonic measure associated to divergence-form uniformly elliptic operators $L = -\operatorname{div}(A\nabla\cdot)$. It is known that fine properties of elliptic measures can differ from classical harmonic measure, e.g. elliptic measure on a smooth domain can be singular with respect to surface measure [MM81], [CFK81], [Wu94]. For $1 \leq \Lambda < \infty$, define the *elliptic Bourgain constant*

$$b_n(\Lambda) := n - \sup_{\Omega, A} \overline{\dim}_H \omega_\Omega^A, \tag{2.4}$$

where the supremum is taken over domains $\Omega \subset \mathbb{R}^n$ and symmetric, measurable, uniformly elliptic matrix fields A satisfying $\Lambda^{-1}|\xi|^2 \leq \langle A(x)\xi, \xi\rangle$ and $\langle A(x)\xi, \eta\rangle \leq \Lambda|\xi||\eta|$ for a.e. x and every $\xi, \eta \in \mathbb{R}^n$. With this normalization, $b_n(1) = b_n$. Examples of Sweezy [Swe92, Swe94] show that $\lim_{\Lambda\to\infty} b_n(\Lambda) = 0$ for every $n \geq 2$. In the plane, Guillén-Mola [GM25] proved the bound $b_2(\Lambda) \geq 2/(\Lambda + 1)$. For $n \geq 3$, it seems likely that positivity of $b_n(\Lambda)$ for each fixed $\Lambda < \infty$ can be obtained by a modification of Bourgain's method, but we are unaware of such a result in the literature.

Challenge 10 (Elliptic Bourgain Challenge)**.** Prove that $b_n(\Lambda) > 0$ for every $n \geq 3$ and every $1 \leq \Lambda < \infty$, and determine the rate at which $b_n(\Lambda) \to 0$ as $\Lambda \to \infty$ for each $n \geq 2$. In particular, obtain quantitative upper and lower bounds for $b_n(\Lambda)$ in terms of Λ.

It would also be interesting to study analogous questions under additional regularity assumptions on the coefficients A or for other classes of operators L.

3. Principles of estimating cubical Hausdorff contents

The purpose of this section is to understand quantitatively when empty space inside a cube makes it advantageous to replace the cube by a cover using smaller cubes. When the dimension parameter is below the ambient dimension, subdivision carries an intrinsic cost, so a hole must be large enough, and the covering cubes chosen efficiently enough, to compensate for that cost. We compare uniform and mixed-scale covers and develop mechanisms for deciding when a hole produces a genuine gain in s-dimensional cubical content. These principles will later be applied to the round holes that arise in Bourgain's alternative. As an indication of the subtleties involved, even the exact cubical Hausdorff content of a Euclidean ball appears to be unknown.

Let $n \geq 1$ be an integer. For the purpose of this paper, a *cube* Q in $\mathbb{R}^n$ of *side length* $0 < \ell < \infty$ is any set of the form $I_1 \times \cdots \times I_n$, where each set I_i is an interval of the form $(a, a+\ell)$, $(a, a+\ell]$, $[a, a+\ell)$, or $[a, a+\ell]$. A singleton $\{a\}$ is not a cube. If Q is a cube, then we may denote its side length by $\operatorname{side} Q$ and its volume by $\operatorname{vol} Q$, i.e. $\operatorname{vol} Q = (\operatorname{side} Q)^n$. Two cubes are said to have the *same type* if their i-th component intervals are of the same type for each $1 \leq i \leq n$. Thus, there are 4^n types of cubes in $\mathbb{R}^n$.

Definition 3.1. Let $s \in [0, \infty)$. For all $\delta \in (0, \infty]$ and for all sets $E \subset \mathbb{R}^n$, we define

$$\mathcal{Q}^s_\delta(E) := \inf \left\{ \sum_1^\infty (\operatorname{side} Q_i)^s : E \subseteq \bigcup_1^\infty Q_i,\ Q_i \text{ is a cube with } \operatorname{side} Q_i \leq \delta \right\}. \tag{3.1}$$

We call $\mathcal{Q}^s_\infty$ the *s-dimensional cubical content* on $\mathbb{R}^n$. The *s-dimensional cubical measure* on $\mathbb{R}^n$ is defined by $\mathcal{Q}^s(E) := \lim_{\delta \downarrow 0} \mathcal{Q}^s_\delta(E)$ for all $E \subset \mathbb{R}^n$.

The quantities $\mathcal{Q}^s_\delta$ and $\mathcal{Q}^s$ are outer measures on $\mathbb{R}^n$ and Borel sets are $\mathcal{Q}^s$ measurable in the sense of Caratheodory. Moreover, $\mathcal{Q}^s_\delta$ and $\mathcal{Q}^s$ share the same null sets. Notably, the contents $\mathcal{Q}^s_\delta$ are *not* Borel measures for any $\delta \in (0, \infty]$ unless $s = n$.

Example 3.2. When $s > n$, $\mathcal{Q}^s_\infty = \mathcal{Q}^s = 0$ (the zero outer measure), and when $s = n$, $\mathcal{Q}^n_\infty = \mathcal{Q}^n = \mathcal{L}^n$ (the Lebesgue outer measure on $\mathbb{R}^n$). When $s < n$, $\mathcal{Q}^s_\infty$ and $\mathcal{Q}^s$ are distinct outer measures. For instance, when $s = 0$, $\mathcal{Q}^0$ is the counting measure and

$$\mathcal{Q}^0_\infty(E) = \begin{cases} 0, & \text{if } E = \emptyset \\ 1, & \text{if } E \text{ is nonempty and bounded} \\ \infty, & \text{if } E \text{ is unbounded} \end{cases}.$$

All sets are $\mathcal{Q}^0$ measurable, but the only $\mathcal{Q}^0_\infty$ measurable sets are $\mathbb{R}^n$ and $\emptyset$.

Principle 3.3 (bigger cubes are more efficient than smaller cubes)**.** Let $s = n - \rho < n$. Suppose that a cube P is partitioned into two or more cubes $Q_1, Q_2, \ldots, Q_k$, i.e. $P = \bigcup_1^k Q_i$ and $Q_i \cap Q_j = \emptyset$ whenever $i \neq j$. While $\operatorname{vol} P = \sum_1^k \operatorname{vol} Q_i$,

$$(\operatorname{side} P)^s = (\operatorname{vol} P)^{1-\rho/n} < (\operatorname{vol} P)^{1-\rho/n} \sum_1^k \left(\frac{\operatorname{vol} Q_i}{\operatorname{vol} P} \right)^{1-\rho/n} = \sum_1^k (\operatorname{side} Q_i)^s,$$

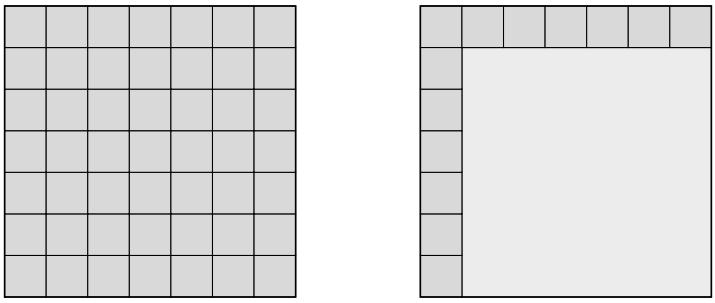


FIGURE 3.1. Illustrating Examples 3.4 and 3.5 with $n = 2$ and $m = 7$. For $s \in (n-1, n)$, the mixed subdivision on the right is substantially more efficient than the regular m-adic subdivision on the left.

because $t_1, \ldots, t_k \in (0,1)$ and $\sum_1^k t_i = 1$ implies $\sum_1^k t_i^{1-\epsilon} > 1$ for all $\epsilon \in (0,1]$. From this we obtain the following simple, but important principle. When using covers of disjoint cubes to estimate s-dimensional cubical contents of sets in $\mathbb{R}^n$ from above, it is generally more efficient to combine smaller cubes into bigger cubes wherever possible.

Example 3.4. Suppose that $s < n$ is fixed. Let $m \geq 3$ be an integer. If we partition a cube P of side length 1 into m^n cubes $Q_1, \ldots, Q_{m^n}$ of side length $1/m$, then

$$\sum_{i=1}^{m^n} (\operatorname{side} Q_i)^s = m^{n-s} \to \infty \quad \text{as } m \to \infty. \tag{3.2}$$

Note that even if we deleted one of the cubes from the list, then it is still possible for the sum over the remaining cubes to exceed 1:

$$S = \sum_{i=1}^{m^n - 1} (\operatorname{side} Q_i)^s = m^{n-s} - m^{-s} > 1$$

unless $n - s$ is sufficiently small depending on m. Thus, if $E \subset Q_1 \cup \cdots \cup Q_{m^n-1}$, then it might still be more efficient to use P to estimate the s-dimensional cubical content of E instead of the smaller cubes $Q_1, \ldots, Q_{m^n-1}$.

Now, $S \leq 1$ if and only if $n - s \leq -\log_m(1 - m^{-n})$. By concavity, the tangent line of $\log(1+x)$ at $x = 0$ lies above the graph of the function. Hence, for all $-1 < x < \infty$, $\log(1+x) \leq x$. It follows that for all $0 \leq x < 1$,

$$x \leq -\log(1-x) = \log\left(\frac{1}{1-x}\right) = \log\left(1 + \frac{x}{1-x}\right) \leq \frac{x}{1-x}. \tag{3.3}$$

Thus, we see that a sufficient condition for $S \leq 1$ is for $n - s \leq 1/(m^n \log(m))$, which roughly speaking is smaller than the volume of the "hole" (the deleted cube) by a factor of $\log(m)$ that blows up as $m \to \infty$.

Example 3.5. Let $s \in (n-1, n)$ be fixed. Let $m \geq 3$ be an integer. If we partition a cube P of side length 1 into one cube Q_0 of side length $1 - (1/m)$ and $N = m^n - (m-1)^n$

cubes $Q_1, \ldots, Q_N$ of side length $1/m$ (see Figure 3.1), then by the binomial theorem,

$$\begin{aligned} \sum_{i=0}^{N} (\operatorname{side} Q_i)^s &= (1-m^{-1})^s + m^{n-s}\left(1-(1-m^{-1})^n\right) \\ &= (1-m^{-1})^s + O(m^{n-s-1}) \to 1 \quad \text{as } m \to \infty, \end{aligned} \tag{3.4}$$

because $n - s - 1 < 0$ and $O(m^{n-s-1}) \le Cm^{n-s-1}$ for some constant C depending only on n. Compare (3.4) with (3.2). This simple observation is one source of our improved estimates for Bourgain's constant b_n compared with the estimates in [BG24].

If we omit one of the smaller cubes, then the s-dimensional content of the list becomes

$$\begin{aligned} \widetilde{S} &= \sum_{i=0}^{N-1} (\operatorname{side} Q_i)^s = (1-m^{-1})^s + m^{n-s}\left(1-(1-m^{-1})^n - m^{-n}\right) \\ &= m^{\rho}\Big(1-(1-(m-1)^{-\rho})(1-m^{-1})^n - m^{-n}\Big), \quad \text{where } \rho = n-s. \end{aligned}$$

Hence $\widetilde{S} \le 1$ if and only if

$$\underbrace{1-m^{-\rho}}_{A} \le \underbrace{(1-(m-1)^{-\rho})}_{B}\underbrace{(1-m^{-1})^n}_{C} + m^{-n}. \tag{3.5}$$

To find sufficient conditions on ρ for $\widetilde{S} \le 1$, let us bound A from above and B and C from below. Applying (3.3) with $x = 1 - m^{-\rho}$, we have

$$A = 1 - m^{-\rho} \le -\log(1-(1-m^{-\rho})) = \rho \log(m).$$

From the Taylor series bound $e^{-x} \le 1 - x + (1/2)x^2$ for all $x \ge 0$, it follows that

$$B = 1 - e^{-\rho \log(m-1)} \ge \rho \log(m-1) - \frac{1}{2}\rho^2 \log(m-1)^2.$$

Next, by considering the sign of the derivative of $f(x) = (1+x)^n - (1+nx)$, one may verify that $f(x)$ has a global minimum on $[-1, \infty)$ at $x = 0$. This is Bernoulli's inequality:

$$(1+x)^n \ge 1 + nx \quad \text{for all } x \ge -1. \tag{3.6}$$

Hence $C = (1-m^{-1})^n \ge 1 - (n/m)$. Substituting our bounds for A, B, and C into (3.5) and rearranging, we discover that $\widetilde{S} \le 1$ if $\rho = n - s < 1$ and ρ satisfies the quadratic inequality

$$\left[\frac{1}{2}\left(1-\frac{n}{m}\right)\log(m-1)^2\right]\rho^2 + \left[\log\left(1+\frac{1}{m-1}\right) + \frac{n}{m}\log(m-1)\right]\rho \le m^{-n}. \tag{3.7}$$

Suppose that $0 < \rho \le m^{-n}/(n+1)$ and recall that $n \ge 1$ and $m \ge 3$. Estimating generously, the left hand side of (3.7) is bounded above by

$$\begin{aligned} &\left[\frac{1}{2}\log(m)^2\right]\frac{m^{-2n}}{(n+1)^2} + \left[\log(3/2) + \frac{n}{m}\log(m)^2\right]\frac{m^{-n}}{n+1} \\ &\qquad \le \left(\frac{1}{2}\log(3/2) + \frac{9}{8}\frac{\log(m)^2}{m}\right)m^{-n} < 0.82m^{-n}, \end{aligned}$$

n	lower(n)	ω_n	upper(n)	n	lower(n)	ω_n	upper(n)
1	1.99851...	2.00000...	2.11268...	6	5.16749...	5.16771...	5.27627...
2	3.14056...	3.14159...	3.27419...	7	4.72462...	4.72476...	4.81293...
3	4.18807...	4.18879...	4.33003...	8	4.05862...	4.05871...	4.12683...
4	4.93430...	4.93480...	5.07329...	9	3.29845...	3.29850...	3.34881...
5	5.26345...	5.26378...	5.39028...	10	2.55013...	2.55016...	2.58579...

FIGURE 3.2. Exact values vs. bounds for ω_n in Lemma 3.7 for $1 \le n \le 10$.

where the reader can verify the final inequality by checking that the expression in round parentheses is maximized over integers $m \ge 3$ when $m = 7$. Thus, a sufficient condition for $\widetilde{S} \le 1$ is for $n - s \le m^{-n}/(n+1)$, which is smaller than the volume of the deleted cube by a factor $n+1$ that is independent of m.

The previous pair of examples illustrate the following principle.

Principle 3.6 (volume of a hole provides a good heuristic for when to use smaller cubes)**.** When there a "hole" or "empty space" Q inside of a cube P, then a cover of $P \setminus Q$ with cubes of side length smaller than P can provide a better estimate for the s-dimensional cubical content of $P \setminus Q$ than (side $P)^s$ as long as $n-s$ is (a bit) smaller than $\mathcal{L}^n(Q)/\mathcal{L}^n(P)$. Using uniform covers of side length t side P requires something like $n - s$ is less than the relative Lebesgue measure divided by $\log(1/t)$. Using covers of mixed sizes can lead to a larger range of valid $n - s$.

Let $B(x, r)$ denote the open ball with center $x \in \mathbb{R}^n$ and radius $r > 0$, respectively.

Lemma 3.7. *Denote the volume of the unit ball in $\mathbb{R}^n$ by ω_n, i.e.*

$$\omega_n = \frac{\pi^{n/2}}{\Gamma((n+2)/2)}, \qquad \Gamma(x) = \int_0^\infty t^{x-1} e^{-t}\, dt \quad (x > 0). \tag{3.8}$$

For all $n \ge 1$,

$$\frac{1}{e^{1/(6n+12)}} \cdot \frac{\sqrt{2\pi e}^{n+2}}{2\sqrt{\pi}^3 \sqrt{n+2}^{n+1}} \le \omega_n \le \frac{\sqrt{2\pi e}^{n+2}}{2\sqrt{\pi}^3 \sqrt{n+2}^{n+1}}. \tag{3.9}$$

Proof. The formulas for the volume of an n-ball are well-known. Apply the Stirling's inequalities for the gamma function, see e.g. [Art64] or [Jam15], with $x = (n+2)/2$:

$$\frac{\sqrt{2\pi}}{\sqrt{x}} \left(\frac{x}{e}\right)^x \le \Gamma(x) \le \frac{\sqrt{2\pi}}{\sqrt{x}} \left(\frac{x}{e}\right)^x e^{1/(12x)} \quad \text{for all } x > 0. \tag{3.10}$$

□

Table (3.2) shows that the bounds in Lemma 3.7 rapidly approximate ω_n, particularly the lower bound with its exponential correction. Note that ω_n is maximized when $n = 5$.

Example 3.8. Let $0 < s < n$. Let's estimate the s-dimensional cubical content of the open unit ball $B(0, 1)$. When $s > n - 1$, it is easy to verify that $\mathcal{Q}^s_\infty(\partial B(0, 1)) = 0$ and

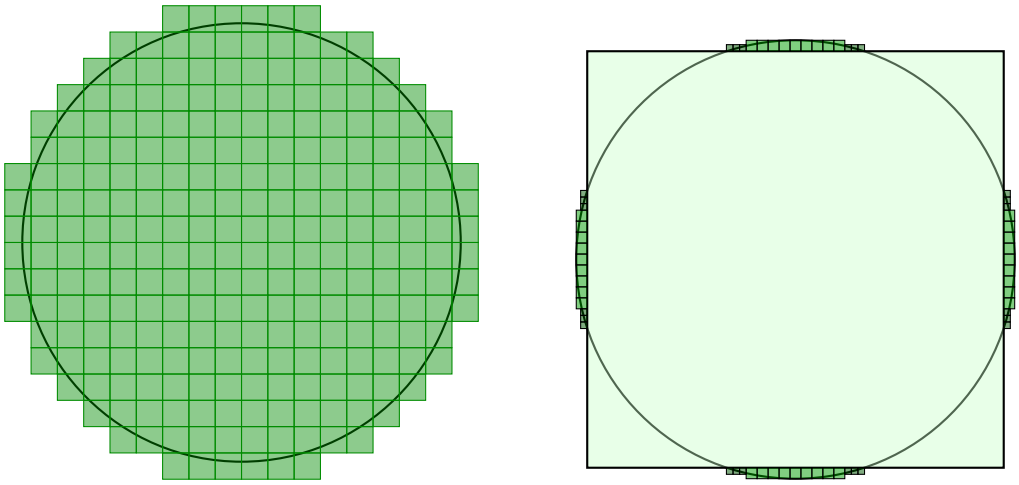


FIGURE 3.3. Illustrating Example 3.8 with $n = 2$ and $s = 1.75$. As $s \uparrow n$, uniform covers require increasingly small cubes, while covers using cubes of several sizes can be more efficient.

$\mathcal{Q}^s_\infty(\overline{B(0,1)}) = \mathcal{Q}^s_\infty(B(0,1))$. Circumscribing $B(0,1)$ with a single cube of side length $2 = \operatorname{diam} B(0,1)$, we obtain an initial bound

$$\mathcal{Q}^s_\infty(B(0,1)) \leq 2^s. \tag{3.11}$$

Cover $B(0,1)$ with pairwise disjoint cubes of side length t that intersect $B(0,1)$; e.g., take cubes from some partition of $\mathbb{R}^n$ by half-open cubes of side length t. Any such cube has diameter $\sqrt{n}t$ and is contained inside $\widehat{B} = B(0, 1+\sqrt{n}t)$. Therefore, the number of cubes N_j of side length t needed to cover $B(0,1)$ is no more than

$$\frac{\text{volume of } \widehat{B}}{\text{volume of a cube}} = \frac{\omega_n(1+\sqrt{n}t)^n}{t^n}.$$

Covering $B(0,1)$ with said cubes, we obtain

$$\mathcal{Q}^s_\infty(B(0,1)) \leq \omega_n(1+\sqrt{n}t)^n t^{-(n-s)}.$$

By a simple calculus exercise, the latter expression is minimized when $t = (n-s)/(s\sqrt{n})$:

$$\mathcal{Q}^s_\infty(B(0,1)) \leq \omega_n \frac{n^n}{s^s}\left(\frac{\sqrt{n}}{n-s}\right)^{n-s}. \tag{3.12}$$

This is a reasonable discovery insofar as the upper bound on $\mathcal{Q}^s_\infty(B(0,1))$ tends to $\omega_n = \mathcal{Q}^n(B(0,1)) = \mathcal{Q}^n_\infty(B(0,1))$ as $s \uparrow n$. However, we note that this bound is not sharp by Principle 3.3, because we could combine many of the little cubes of side length t inside of $B(0,1)$ into bigger cubes. To the authors' knowledge, the exact value of the content $\omega_{n,s} := \mathcal{Q}^s_\infty(B(0,1))$ is unknown and computing it is an *open problem*.

For instance, when $n = 2$ and $s = 1.75$, we can cover $B(0,1)$ with 256 squares of side length $t = 0.12$ (see Figure 3.3); this cover leads to the bound $\mathcal{Q}^{1.75}_\infty(B(0,1)) \leq 6.2633...$, which is substantially worse than the bound $\mathcal{Q}^{1.75}_\infty(B(0,1)) \leq 2^{1.75} = 3.3635...$ from (3.11).

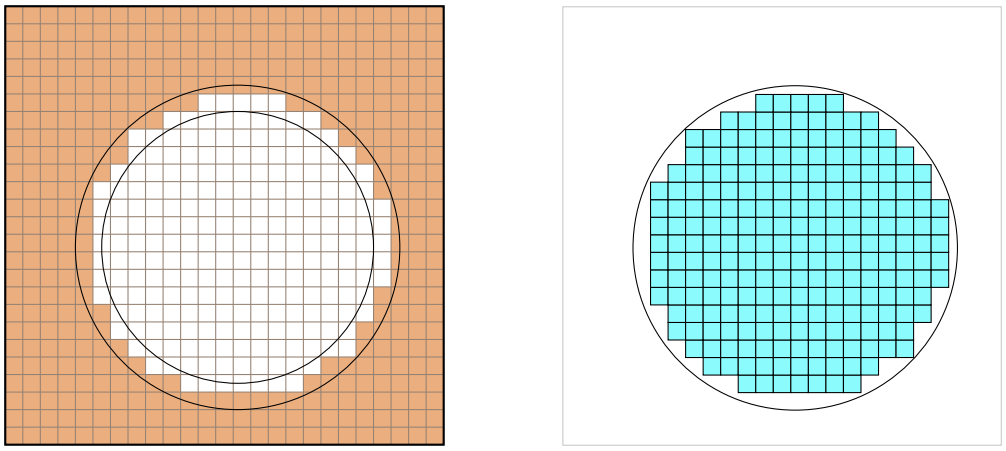


FIGURE 3.4. Illustrating Example 3.9 with $n = 2$ and $h = 25$. The blue squares lie inside $B(x, r)$, while the orange squares intersect the complement of $B(x, r)$ and are disjoint from the interior of $B(x, t)$.

However, we can beat the $2^{1.75}$ bound by using several side lengths: cover $B(0, 1)$ with one cube of side length 1.9, 36 cubes of side length 0.05 and 24 cubes of side length 0.03 to obtain $\mathcal{Q}_\infty^{1.75}(B(0, 1)) \leq 3.3170...$, greater than 1% improvement over (3.11).

Example 3.9. Let $s \in (0, n)$, close to n, and let $B = B(x, r)$ be any (open) ball contained in $[0, 1]^n$. We wish to estimate the s-dimensional cubical content of the set $E = [0, 1]^n \setminus B$ in terms of n, s, and r, but independent of the location of B within the $[0, 1]^n$. In contrast with the previous example, we add a further stipulation: we want to estimate $\mathcal{Q}_\infty^s(E)$ using only pairwise disjoint cubes that are contained in $[0, 1]^n$.

Let $h \geq 2$ be a (large) integer. Partition $[0, 1]^n$ into a family $\mathcal{Q}_h$ of h^n pairwise disjoint cubes of side length $1/h$. Let $\mathcal{R}_h$ denote the subfamily of all cubes in $\mathcal{Q}_h$ that intersect E. Given $0 < t < r$ (close to r), assume that h is sufficiently large so that $r - \sqrt{n}/h \geq t$, i.e. assume that $h \geq \sqrt{n}/(r(1 - t/r))$. We remark that h tends to infinity as the radius of B goes to 0 and as well as when t approaches r. No cube in $\mathcal{R}_h$ intersects the interior of $B(x, t)$. Thus,

$$\#\mathcal{R}_h \leq \frac{\text{volume of } [0, 1]^n \,-\, \text{volume of } B(x, t)}{\text{volume of a cube}} = \frac{1 - \omega_n t^n}{(1/h)^n}.$$

Because the cubes in $\mathcal{R}_h$ cover E and have side length $1/h$, we obtain

$$\mathcal{Q}_\infty^s(E) \leq \sum_{R \in \mathcal{R}_h} (\text{side } R)^s \leq h^{n-s} \left(1 - \omega_n t^n\right). \tag{3.13}$$

In order for (3.13) to improve on the trivial bound $\mathcal{Q}_\infty^s(E) \leq \mathcal{Q}_\infty^s([0, 1]^n) \leq 1$, we need s to be sufficiently close to n so that $(n - s)\log(h) < -\log\left(1 - \omega_n t^n\right)$. In particular, it suffices to take $n - s < \omega_n t^n/(\log h)$, which is consistent with Principle 3.6.

4. Bourgain-type dimension lemma with cubes in general position

For $s \in [0,\infty)$ and $E \subset \mathbb{R}^n$, let $\mathcal{H}^s(E)$ denote the s-dimensional Hausdorff measure, defined as in Definition 3.1 but allowing covers by arbitrary sets $E_i \subset \mathbb{R}^n$ and replacing $(\operatorname{side} Q_i)^s$ by $(\operatorname{diam} E_i)^s$. Since $\mathcal{Q}^s$ and $\mathcal{H}^s$ are comparable up to dimensional constants,

$$\dim_H E := \inf\{s \geq 0 : \mathcal{H}^s(E) < \infty\} = \inf\{s \geq 0 : \mathcal{Q}^s(E) < \infty\}.$$

For a Borel measure μ on $\mathbb{R}^n$, its *upper Hausdorff dimension* is

$$\overline{\dim}_H \mu := \inf\{\dim_H E : \mu(\mathbb{R}^n \setminus E) = 0,\ E \text{ Borel}\}.$$

We will use the countable stability of both notions: $\dim_H \bigcup_i E_i = \sup_i \dim_H E_i$, and if $\mathbb{R}^n = \bigcup_i E_i$ with each E_i Borel, then $\overline{\dim}_H, \mu = \sup_i \overline{\dim}_H(\mu \llcorner E_i)$.

The following dimension lemma originates in Bourgain's work on the dimension of harmonic measure [Bou87], where the relevant estimates appear within the proof in a form adapated to its specific setting. An explicit formulation and proof appear in the parabolic setting in [BG25, Theorem 2.10] and a Euclidean version appears in [BG24]. The formulation below allows pairwise disjoint covering cubes in general position and arbitrary $\alpha \in (0,1)$. We also refine the proof to eliminate a loss in the type (II) iteration, thereby recovering the factor $1-\alpha$ in the definition of λ_α.

Theorem 4.1 (Bourgain-type dimension lemma with cubes in general position)**.** *Let μ be a Borel regular outer measure on $\mathbb{R}^n$ that is finite on bounded sets, and let $E \subset \mathbb{R}^n$ be a Borel set with $\mu(\mathbb{R}^n \setminus E) = 0$. Suppose that there exist parameters $\alpha \in (0,1)$, $\rho \in (0,n)$, $\beta \in (0,1)$, $\tau \in (1,\infty)$, and $\gamma \in (0,1)$ such that for every cube P in $\mathbb{R}^n$ with $\operatorname{side} P \leq 1$, at least one (possibly both) of the following conditions holds:*

(I) *there is a cover of $E \cap P$ by a finite or countable list of pairwise disjoint cubes $Q_1, Q_2, \ldots$ contained in P with $\operatorname{side} Q_i \leq \beta \operatorname{side} P$ such that*

$$\sum_i (\operatorname{side} Q_i)^{n-\rho} \leq (\operatorname{side} P)^{n-\rho};$$

(II) *there is a cover of $E \cap P$ by a finite or countable list of pairwise disjoint cubes $R_1, R_2, \ldots$ contained in P with $\operatorname{side} R_i \leq \beta \operatorname{side} P$ such that*

$$\sum_i (\operatorname{side} R_i)^{n-\rho} \leq \tau\,(\operatorname{side} P)^{n-\rho} \quad \text{and} \quad \sum_i \mu(R_i)^\alpha (\operatorname{vol} R_i)^{1-\alpha} \leq \gamma\,\mu(P)^\alpha (\operatorname{vol} P)^{1-\alpha}.$$

Then the upper Hausdorff dimension of μ is at most

$$n - \left(\frac{\lambda_\alpha}{1+\lambda_\alpha}\right)\rho, \quad \text{where } \lambda_\alpha := \frac{-\log_\tau(\gamma)}{1-\alpha} \in (0,\infty). \tag{4.1}$$

One may, if desired, restrict throughout to cubes that are half-open of some fixed type.

Proof. We may assume that $\mu(E) > 0$, otherwise μ is just the zero measure. Although the theorem permits cubes of different types (for later applications), the proof uses only their nesting, pairwise disjointness, and side lengths, not the assignment of cube boundaries. Thus, if the hypotheses are known only for half-open cubes of one fixed type, the same

proof applies with the initial unit cube chosen to have that type. By countable stability, it suffices to restrict μ to a unit cube P_0 of positive measure. Replacing E by $E \cap P_0$ and μ by $\mu \llcorner P_0$, and then normalizing μ, we may assume that $E \subset P_0$, $\operatorname{side} P_0 = 1$, and $\mu(P_0) = 1$. Conditions (I) and (II) persist under these reductions.

Fix $0 < \epsilon < 1$ and $0 < \sigma < \lambda_\alpha \rho \epsilon$. We will prove that

$$\overline{\dim}_H \mu \leq \max\{n - \rho(1-\epsilon), n - \sigma\}.$$

This suffices, because for $\epsilon = 1/(1+\lambda_\alpha)$, we have $\rho(1-\epsilon) = \lambda_\alpha \rho \epsilon = \lambda_\alpha \rho/(1+\lambda_\alpha)$, and σ may be taken arbitrarily close to $\lambda_\alpha \rho \epsilon$ from below. Fix any $0 < \delta < 1$ sufficiently small such that $\log_\tau(\delta^{-\rho\epsilon}) \geq 2$ and let $s = \lceil \log_\tau(\delta^{-\rho\epsilon}) \rceil$. Since $s - 1 < \log_\tau(\delta^{-\rho\epsilon}) \leq s$,

$$\tau^{s-1} < \delta^{-\rho\epsilon} \leq \tau^s. \tag{4.2}$$

We now inductively build a tree $\mathcal{T} = \bigcup_{l=0}^{L} \mathcal{T}_l$ of finite depth L. Each set in $\mathcal{T}$ is a cube contained in P_0 and the cubes in any given level $\mathcal{T}_l$ are pairwise disjoint. Each cube $Q \in \mathcal{T}_{l+1}$ is contained in a unique cube $P \in \mathcal{T}_l$ and satisfies $\operatorname{side} Q \leq \beta \operatorname{side} P$, but in general there is no lower bound on $\operatorname{side} Q$ in terms of $\operatorname{side} P$. For any cube $P \in \mathcal{T}_l$, we let $\mathsf{Child}(P) = \{Q \in \mathcal{T}_{l+1} : Q \subset P\}$. We call $P \in \mathcal{T}$ *terminal* if $\operatorname{side} P \leq \delta$ or $E \cap P = \emptyset$. Terminal cubes will have no children, while every non-terminal cube will have at least one child. The depth L is finite, because $\operatorname{side} Q \leq \beta \operatorname{side} P$ forces the side lengths along any chain to decay geometrically until the terminal condition $\operatorname{side} P \leq \delta$ is reached.

Base Case. At the top level, put $\mathcal{T}_0 = \{P_0\}$.

Induction Step. Suppose that $\mathcal{T}_l \neq \emptyset$ for some $l \geq 0$ and let $P \in \mathcal{T}_l$. If P is terminal, we add no children. Otherwise, we shall include a nonempty, finite or countable family of pairwise disjoint subcubes of P as sets in $\mathcal{T}_{l+1}$. On the one hand, if P satisfies condition (I), then we say that P is *type 1*. Choose pairwise disjoint cubes $Q_1, Q_2, \ldots$ contained in P with $\operatorname{side} Q_i \leq \beta \operatorname{side} P$ that cover $E \cap P$ and have

$$\sum_i (\operatorname{side} Q_i)^{n-\rho} \leq (\operatorname{side} P)^{n-\rho}. \tag{4.3}$$

In this case, we include the cube $Q_i \in \mathcal{T}_{l+1}$ for each i, so that $\mathsf{Child}(P) = \{Q_1, Q_2, \ldots\}$. On the other hand, if P fails condition (I), then P satisfies condition (II) and we say that P is *type 2*. Choose pairwise disjoint cubes $R_1, R_2, \ldots$ contained in P with $\operatorname{side} R_i \leq \beta \operatorname{side} P$ that cover $E \cap P$ and satisfy

$$\sum_i (\operatorname{side} R_i)^{n-\rho} \leq \tau \, (\operatorname{side} P)^{n-\rho}, \quad \sum_i \mu(R_i)^\alpha (\operatorname{vol} R_i)^{1-\alpha} \leq \gamma \, \mu(P)^\alpha (\operatorname{vol} P)^{1-\alpha}. \tag{4.4}$$

In this case, we include each R_i in $\mathcal{T}_{l+1}$, so that $\mathsf{Child}(P) = \{R_1, R_2, \ldots\}$. Repeating this procedure for each $P \in \mathcal{T}_l$ completes the definition of $\mathcal{T}_{l+1}$.

Goal. Write $\eta := \max\{n - \rho(1-\epsilon), n - \sigma\}$. We aim to find a Borel set $F_\delta \subset E$ with

$$\lim_{\delta \downarrow 0} \mu(\mathbb{R}^n \setminus F_\delta) = \lim_{\delta \downarrow 0} \mu(E \setminus F_\delta) = 0 \quad \text{and} \quad \mathcal{Q}^\eta_{\zeta(\delta)}(F_\delta) \leq 2 < \infty, \tag{4.5}$$

where $\zeta(\delta) \to 0$ as $\delta \to 0$ is a scale to be specified below.

Let $\mathcal{T}^*$ denote the terminal cubes in $\mathcal{T}$ that intersect E. Every $Q \in \mathcal{T}^*$ satisfies $\operatorname{side} Q \le \delta$. Since $\operatorname{side} P_0 = 1 > \delta$ and $\mu(E) = 1$, the initial cube P_0 is not terminal. We now use terminal cubes to identify two collections $\mathcal{E}, \mathcal{F} \subset \mathcal{T}$ with $\mathcal{E} \cap \mathcal{F} = \emptyset$, as well as a subfamily $\mathcal{G} \subset \mathcal{F}$. We say that $Q \in \mathcal{T}^* \cap \mathcal{T}_l$ is *efficient for* $\mathcal{Q}^{n-\rho}$ and include $Q \in \mathcal{E}$ if the chain $P_0, \dots, P_{l-1}$ of ancestors of Q contains fewer than s type 2 cubes. Otherwise, there is a unique ancestor P of Q that is the s-th type 2 cube in the chain, and we include $P \in \mathcal{F}$. By construction,

$$E \subset \bigcup \mathcal{T}^* \subset \bigcup \mathcal{E} \cup \bigcup \mathcal{F}.$$

Furthermore, $\operatorname{side} Q \le \delta$ for all $Q \in \mathcal{E}$ and $\delta < \operatorname{side} Q \le \beta^{s-1}$ for all $Q \in \mathcal{F}$, since cubes in $\mathcal{E}$ belong to $\mathcal{T}^*$ and cubes in $\mathcal{F}$ are not terminal and have at least $s-1$ ancestors. Thus, $\mathcal{E} \cup \mathcal{F}$ covers E by cubes of side length at most $\zeta(\delta) := \max\{\delta, \beta^{s-1}\}$. With ϵ fixed, as $\delta \to 0$ we have $s \to \infty$ and $\zeta(\delta) \to 0$. Finally, let $\mathcal{G}$ consist of those $Q \in \mathcal{F}$ satisfying

$$(\operatorname{side} Q)^{n-\sigma} \le \mu(Q)$$

and put $F_\delta = E \cap \bigcup(\mathcal{E} \cup \mathcal{G})$. Since E is Borel, the set F_δ is Borel.

We now make estimates towards (4.5). Both of the sums below run over cubes lying at different levels of $\mathcal{T}$ and are controlled by the following observation. For $Q \in \mathcal{T}$, let $t(Q)$ denote the number of ancestors of Q in $\mathcal{T}$ of type 2. Suppose that a number $w(Q) \ge 0$ is assigned to each $Q \in \mathcal{T}$ and $\kappa > 0$ is a constant such that

$$\sum_{Q \in \mathsf{Child}(P)} w(Q) \le w(P) \text{ if } P \text{ is type 1}, \qquad \sum_{Q \in \mathsf{Child}(P)} w(Q) \le \kappa w(P) \text{ if } P \text{ is type 2}. \tag{4.6}$$

If $\mathcal{A} \subset \mathcal{T}$ is a pairwise disjoint family of cubes, then

$$\sum_{Q \in \mathcal{A}} w(Q) \le \Big(\sup_{Q \in \mathcal{A}} \kappa^{t(Q)} \Big) w(P_0). \tag{4.7}$$

Indeed, put $W(P) = \sum\{w(Q) : Q \in \mathcal{A},\, Q \subseteq P\}$ and $M(P) = \sup\{\kappa^{t(Q)-t(P)} : Q \in \mathcal{A},\, Q \subseteq P\}$. A downward induction gives $W(P) \le M(P) w(P)$: this is immediate if no cube of $\mathcal{A}$ lies in P or if $P \in \mathcal{A}$, and otherwise follows by summing over $\mathsf{Child}(P)$ and applying (4.6). Taking $P = P_0$, for which $t(P_0) = 0$, gives (4.7).

Assign $w(Q) = (\operatorname{side} Q)^{n-\rho}$ and $\kappa = \tau$, so that (4.6) is precisely (4.3) together with the first inequality in (4.4). Cubes in $\mathcal{E}$ are terminal, hence pairwise disjoint, and $t(Q) \le s-1$ for each of them. Thus, by (4.7) and (4.2),

$$\sum_{Q \in \mathcal{E}} (\operatorname{side} Q)^{n-\rho} \le \tau^{s-1} \le \delta^{-\rho\epsilon}.$$

On the one hand, since $\operatorname{side} Q \le \delta < 1$ for all $Q \in \mathcal{E}$,

$$\mathcal{Q}^\eta_\delta(E \cap \bigcup \mathcal{E}) \le \sum_{Q \in \mathcal{E}} (\operatorname{side} Q)^{n-\rho(1-\epsilon)} \le \delta^{\rho\epsilon} \sum_{Q \in \mathcal{E}} (\operatorname{side} Q)^{n-\rho} \le 1.$$

On the other hand, since $\operatorname{side} Q \le \beta^{s-1} < 1$ for all $Q \in \mathcal{G}$,

$$\mathcal{Q}^\eta_{\beta^{s-1}}(E \cap \bigcup \mathcal{G}) \le \sum_{Q \in \mathcal{G}} (\operatorname{side} Q)^{n-\sigma} \le \sum_{Q \in \mathcal{G}} \mu(Q) \le 1,$$

because cubes in $\mathcal{G}$ are pairwise disjoint subsets of P_0, satisfy the defining inequality for $\mathcal{G}$, and $\mu(P_0) = 1$. Thus,

$$\mathcal{Q}^\eta_{\zeta(\delta)}(F_\delta) \le \mathcal{Q}^\eta_\delta(E \cap \bigcup \mathcal{E}) + \mathcal{Q}^\eta_{\beta^{s-1}}(E \cap \bigcup \mathcal{G}) \le 2 < \infty.$$

To verify (4.5), it remains to estimate $\mu(E \setminus F_\delta)$ in terms of δ.

For every $Q \in \mathcal{F} \setminus \mathcal{G}$, we have

$$\operatorname{vol} Q = (\operatorname{side} Q)^\sigma (\operatorname{side} Q)^{n-\sigma} > \delta^\sigma \mu(Q). \tag{4.8}$$

Furthermore, by Hölder's inequality with exponents $1/\alpha$ and $1/(1-\alpha)$, for all non-terminal cubes $P \in \mathcal{T}$,

$$\begin{aligned} \sum_{Q \in \mathsf{Child}(P)} \mu(Q)^\alpha (\operatorname{vol} Q)^{1-\alpha} &\le \biggl(\sum_{Q \in \mathsf{Child}(P)} \mu(Q) \biggr)^\alpha \biggl(\sum_{Q \in \mathsf{Child}(P)} \operatorname{vol} Q \biggr)^{1-\alpha} \\ &\le \mu(P)^\alpha (\operatorname{vol} P)^{1-\alpha}, \end{aligned} \tag{4.9}$$

because the children of a cube are pairwise disjoint subcubes. Now assign $w(Q) = \mu(Q)^\alpha (\operatorname{vol} Q)^{1-\alpha}$ and $\kappa = \gamma$, so that (4.6) is (4.9) together with the second inequality in (4.4). The cubes in $\mathcal{F}$ are pairwise disjoint and $t(Q) = s - 1$ for each of them, whence (4.7) and $\mu(P_0) = \operatorname{vol}(P_0) = 1$ yield

$$\sum_{Q \in \mathcal{F}} \mu(Q)^\alpha (\operatorname{vol} Q)^{1-\alpha} \le \gamma^{s-1}.$$

Combining these estimates gives

$$\begin{aligned} \mu(E \setminus F_\delta) \le \sum_{Q \in \mathcal{F} \setminus \mathcal{G}} \mu(Q) = \sum_{Q \in \mathcal{F} \setminus \mathcal{G}} \mu(Q)^\alpha \mu(Q)^{1-\alpha} &\overset{(4.8)}{<} \delta^{-\sigma(1-\alpha)} \sum_{Q \in \mathcal{F} \setminus \mathcal{G}} \mu(Q)^\alpha (\operatorname{vol} Q)^{1-\alpha} \\ &\le \delta^{-\sigma(1-\alpha)} \gamma^{s-1}. \end{aligned}$$

Since $\gamma = \tau^{-(1-\alpha)\lambda_\alpha}$ and $\tau^{-s} \le \delta^{\rho\epsilon}$ by (4.2),

$$\gamma^{s-1} = \tau^{(1-\alpha)\lambda_\alpha} (\tau^{-s})^{(1-\alpha)\lambda_\alpha} \le \tau^{(1-\alpha)\lambda_\alpha} \delta^{(1-\alpha)\lambda_\alpha \rho\epsilon}.$$

Combining the last two estimates,

$$\mu(E \setminus F_\delta) \le \tau^{(1-\alpha)\lambda_\alpha} \delta^{(1-\alpha)(\lambda_\alpha \rho\epsilon - \sigma)} \to 0 \quad \text{as } \delta \to 0, \tag{4.10}$$

because $\sigma < \lambda_\alpha \rho\epsilon$. We have verified (4.5).

Conclusion. By (4.10), we may choose $\delta_1 > \delta_2 > \cdots \to 0$ such that $\sum_1^\infty \mu(E \setminus F_{\delta_j}) < \infty$ and $\log_\tau(\delta_1^{-\rho\epsilon}) \ge 2$, and put $F := \liminf_{j\to\infty} F_{\delta_j} = \bigcup_{j=1}^\infty \bigcap_{i=j}^\infty F_{\delta_i}$. By the Borel–Cantelli lemma, $E \setminus F = \limsup_{j\to\infty} (E \setminus F_{\delta_j})$ has μ measure zero. Finally, F is Borel and

$$\mathcal{Q}^\eta(F) = \lim_{j\to\infty} \mathcal{Q}^\eta \left(\bigcap_{i=j}^\infty F_{\delta_i} \right) = \lim_{j\to\infty} \sup_{k \ge j} \mathcal{Q}^\eta_{\zeta(\delta_k)} \left(\bigcap_{i=j}^\infty F_{\delta_i} \right) \le 2 < \infty$$

by continuity from below and (4.5). Therefore, $\overline{\dim}_H \mu \le \dim_H F \le \eta$. □

5. Frostman measures and approximate polar integration

5.1. **Frostman's lemma with constant 1.** In this paper, in order to describe the local size of a set using disjoint coverings of cubes, with both flexible positioning and enough rigidity for computational ease, we utilize the following terminology.

Definition 5.1. Let $n \geq 1$ and $m \geq 2$ be integers. Let $t \in (0, \infty)$. An *m-adic net Δ at initial scale t* is a family of cubes in $\mathbb{R}^n$ such that

(i) Every cube $Q \in \Delta$ has side length tm^{-k} for some integer $k \geq 0$.
(ii) The set Δ_k of cubes in Δ of side length tm^{-k} is pairwise disjoint.
(iii) For every integer $k \geq 0$ and every cube $Q \in \Delta_k$, there exists a family $\mathsf{Child}(Q) \subseteq \Delta_{k+1}$ such that $Q = \bigcup \mathsf{Child}(Q)$; moreover, $\Delta_{k+1} = \bigcup_{Q \in \Delta_k} \mathsf{Child}(Q)$.

We call the set Δ_k the *k-th level* of Δ. We call the set $\bigcup \Delta$ the *site* of Δ and denote it by $\mathsf{Site}(\Delta)$. A 2-adic net is called a *dyadic net*.

Note that we do not require the cubes in an m-adic net Δ to be of any particular type. Furthermore, each level of a net partitions the site of the net. We introduced the concept of the site of a net in order to localize certain estimates; e.g., see §5.2 and (6.7). In the latter application, the site of the net $\Delta_{P,x}$ is assembled from small cubes in a multiscale partition of a cube P that fill out a ball from inside. Net content then permits these top-level cubes to be refined at smaller scales to cover $\partial\Omega$ efficiently within the site.

Definition 5.2. Let Δ be an m-adic net and let $s \in [0, \infty)$. For every set $E \subset \mathbb{R}^n$, we define its *s-dimensional Δ content* by

$$\mathcal{M}_\infty^{\Delta,s}(E) := \inf \left\{ \sum_1^\infty (\operatorname{side} Q_i)^s : E \cap \mathsf{Site}(\Delta) \subseteq \bigcup_1^\infty Q_i,\ Q_i \in \Delta \right\}. \tag{5.1}$$

The quantity $\mathcal{M}_\infty^{\Delta,s}$ is an outer measure on $\mathbb{R}^n$ that is carried by the site of the net in the sense that $\mathcal{M}_\infty^{\Delta,s}(\mathbb{R}^n \setminus \mathsf{Site}(\Delta)) = 0$. The main advantage of net contents over cubical contents lies in our ability to restrict to disjoint covers. The main disadvantage is that net contents are neither translation nor dilation invariant.

Lemma 5.3. *Let Δ be an m-adic net in $\mathbb{R}^n$ and let $s \in [0, \infty)$. For every set $E \subset \mathbb{R}^n$,*

$$\mathcal{M}_\infty^{\Delta,s}(E) = \inf \left\{ \sum_i (\operatorname{side} Q_i)^s : \begin{array}{l} Q_1, Q_2, \cdots \in \Delta \text{ are pairwise disjoint, and} \\ E \cap \mathsf{Site}(\Delta) \subseteq \bigcup_i Q_i \subseteq \mathsf{Site}(\Delta) \end{array} \right\}.$$

Proof. Each family on the right is admissible in the definition of $\mathcal{M}_\infty^{\Delta,s}(E)$, whence the right hand side is at least $\mathcal{M}_\infty^{\Delta,s}(E)$. Conversely, let $Q_1, Q_2, \cdots \in \Delta$ cover $E \cap \mathsf{Site}(\Delta)$. If Q_j has side length tm^{-k}, then any cube in the family that contains Q_j has side length $tm^{-\ell}$ for some integer $0 \leq \ell \leq k$. Thus, each Q_j is contained in a cube of the family that is maximal with respect to inclusion. Let I index the distinct maximal cubes. Two cubes in Δ are either disjoint or nested, so $\{Q_i\}_{i \in I}$ is pairwise disjoint and covers $E \cap \mathsf{Site}(\Delta)$. Since the maximal cubes form a subfamily of the original family, $\sum_{i \in I} (\operatorname{side} Q_i)^s \leq \sum_i (\operatorname{side} Q_i)^s$. $\square$

We use the following variant of Frostman's lemma for dimensions close to n. Without the assumption $s > n-1$, the lemma remains valid, but the bound $\mu(K) \geq \mathcal{M}_\infty^{\Delta,s}(K)$ must be replaced by $\mu(K) \geq 2^{-n}\mathcal{M}_\infty^{\Delta,s}(K)$.

Lemma 5.4 ([BG24, Lemma 2.3])**.** *Let Δ be any m-adic net in $\mathbb{R}^n$. If K is a compact set with $K \subseteq \mathsf{Site}(\Delta)$ and $s \in (n-1, n]$, then there exists a Radon measure μ on $\mathbb{R}^n$ such that $\mu(Q) \leq (\operatorname{side} Q)^s$ for all $Q \in \Delta$ and $\mu(\mathbb{R}^n) = \mu(K) \geq \mathcal{M}_\infty^{\Delta,s}(K)$.*

5.2. **Approximate polar integration of radial functions.** In this subsection, we let Δ denote any fixed dyadic net with initial scale $h^{-1} \in (0, 1]$ for some integer $h \geq 1$. For convenience, let $B^\Delta(x, r) := B(x, r) \cap \mathsf{Site}(\Delta)$ denote the portion of the open ball $B(x, r)$ lying in the site of the net.

Definition 5.5. Let $n \geq 3$. For all $s \in (n-1, n]$ and $z \in \mathbb{R}^n$, define the constant

$$O_z(n, s; \Delta) := \sup_\mu \int_{B^\Delta(0,1/2)} \frac{1}{|x - z|^{n-2}} d\mu(x), \tag{5.2}$$

where the supremum is over all Borel measures μ on $\mathbb{R}^n$ carried by $B^\Delta(0, 1/2)$, in the sense that $\mu(\mathbb{R}^n \setminus B^\Delta(0, 1/2)) = 0$, and satisfying the Frostman-type bound $\mu(Q) \leq (\operatorname{side} Q)^s$ for all $Q \in \Delta$. Define

$$O(n, s; \Delta) := \sup_{z \in \mathbb{R}^n} O_z(n, s; \Delta). \tag{5.3}$$

Lemma 5.6. *Let $n \geq 3$ and $s \in (n-1, n]$. If $B(0, 1/2) \subset \mathsf{Site}(\Delta)$, then*

$$O(n, s; \Delta) \geq O_0(n, s; \Delta) \geq \int_{B(0,1/2)} \frac{1}{|x|^{n-2}}\, dx = \frac{n\omega_n}{8}.$$

Proof. The first inequality is apparent. If μ is Lebesgue measure restricted to $B(0, 1/2)$, then $\mu(Q) \leq (\operatorname{side} Q)^n \leq (\operatorname{side} Q)^s$ for all $Q \in \Delta$, because cubes in Δ have side length at most $h^{-1} \leq 1$. Thus, we obtain a lower bound on $O_0(n, s; \Delta)$ by taking $z = 0$ and integrating $\int_{B(0,1/2)} |x|^{-(n-2)}\, dx$ using polar integration (recall that $n\omega_n$ is the surface measure of the unit sphere $S^{n-1} \subset \mathbb{R}^n$). □

To estimate the dimension of harmonic measure using Bourgain's method, we need to upper bound $O(n, s; \Delta)$ when s is close to n. We achieve this through a flexible scheme to approximate integrals of radial functions against Frostman-type measures. The main idea is to use Whitney-type cubes to transfer part of the homogeneity from the Frostman condition to the integrand in order to convert the original integral into polar integration against Lebesgue measure. The details will take up the remainder of this subsection.

Lemma 5.7. *Let $n \geq 3$ and $h \geq n^3$. For all $z \in B(0, 2)$ and $0 < \xi \leq (3h)^{-1}$, there exists a family of pairwise disjoint cubes $\mathcal{W} \subset \Delta$ such that*

$$B^\Delta(0, 1/2) \setminus \{z\} \subset \bigcup \mathcal{W} \subset B^\Delta(0, 1/2 + 2.5\sqrt{n}\xi), \tag{5.4}$$

$$0 < \frac{\xi \operatorname{dist}(z, Q)}{2(1 + \sqrt{n}\xi)} < \operatorname{side} Q \leq \xi \operatorname{dist}(z, Q) \quad \textit{for all } Q \in \mathcal{W}. \tag{5.5}$$

Proof. Let z and ξ be given. If $B^\Delta(0,1/2)\setminus\{z\}=\emptyset$, then the lemma holds with $\mathcal{W}=\emptyset$. Thus, let us assume that $B^\Delta(0,1/2)\setminus\{z\}$ is nonempty. Say that a cube $Q\in\Delta$ is *admissible* if (i) $Q\cap B^\Delta(0,1/2)\neq\emptyset$, (ii) $\operatorname{dist}(z,Q)>0$, and (iii) $\operatorname{side} Q\le\xi\operatorname{dist}(z,Q)$. Every admissible cube satisfies

$$\operatorname{side} Q\le\xi\operatorname{dist}(z,Q)\le\xi\left(|z|+\tfrac12\right)\le\frac{5}{6h}<h^{-1}, \tag{5.6}$$

since $Q\cap B^\Delta(0,1/2)\neq\emptyset$, $z\in B(0,2)$, and $\xi\le(3h)^{-1}$. In particular, since Δ has initial scale h^{-1}, every admissible cube Q has a unique parent $Q^\uparrow\in\Delta$.

Let $\mathcal{W}$ denote the family of admissible cubes that are maximal with respect to inclusion among admissible cubes. Then $\mathcal{W}$ is pairwise disjoint, because distinct cubes in Δ are disjoint or nested. If $x\in B^\Delta(0,1/2)\setminus\{z\}$, then every sufficiently small cube $Q\in\Delta$ containing x is admissible, since $\operatorname{dist}(z,Q)\to|x-z|>0$ and $\operatorname{side} Q\to0$ as $\operatorname{side} Q\downarrow0$; hence $\bigcup\mathcal{W}\supset B^\Delta(0,1/2)\setminus\{z\}$. Similarly, since $\operatorname{diam} Q=\sqrt{n}\operatorname{side} Q\le2.5\sqrt{n}\xi$ for every $Q\in\mathcal{W}$ by (5.6), we have $\bigcup\mathcal{W}\subset B^\Delta(0,1/2+2.5\sqrt{n}\xi)$.

Claim: $\operatorname{dist}(z,Q^\uparrow)>0$ *for every* $Q\in\mathcal{W}$. Assume to the contrary that $\operatorname{dist}(z,Q^\uparrow)=0$ for some $Q\in\mathcal{W}$. Then, since $\xi\le(3h)^{-1}\le(3n^3)^{-1}\le(6\sqrt{n})^{-1}$ and $\operatorname{side} Q\le\xi\operatorname{dist}(z,Q)$,

$$\operatorname{dist}(z,Q)\le\operatorname{dist}(z,Q^\uparrow)+\operatorname{diam}Q^\uparrow=2\sqrt{n}\operatorname{side}Q\le\frac{1}{3\xi}\operatorname{side}Q\le(1/3)\operatorname{dist}(z,Q).$$

Since $\operatorname{dist}(z,Q)>0$, this is impossible. The claim is established.

It remains to check the lower bound in (5.5). To that end, fix $Q\in\mathcal{W}$. The parent $Q^\uparrow$ satisfies (i), because $Q\subset Q^\uparrow$, and satisfies (ii), by the Claim. Thus, maximality of Q forces $Q^\uparrow$ to fail (iii); that is, $\operatorname{side}Q^\uparrow>\xi\operatorname{dist}(z,Q^\uparrow)$. Hence

$$\begin{aligned}2\operatorname{side}Q=\operatorname{side}Q^\uparrow&>\xi\operatorname{dist}(z,Q^\uparrow)\ge\xi\left(\operatorname{dist}(z,Q)-\operatorname{diam}Q^\uparrow\right)\\&=\xi\operatorname{dist}(z,Q)-2\sqrt{n}\xi\operatorname{side}Q,\end{aligned}$$

and collecting the terms in $\operatorname{side}Q$ establishes (5.5). □

Lemma 5.8. *Let $n\ge3$, let $s\in(n-1,n]$, and let μ be any admissible measure in the definition of $O(n,s;\Delta)$. Let $h\ge n^3$. Given $z\in B(0,2)$ and $0<\xi\le(3h)^{-1}$, let $\mathcal{W}$ be a family of cubes as in Lemma 5.7. For all $Q\in\mathcal{W}$,*

$$\int_Q\frac{1}{|x-z|^{n-2}}\,d\mu(x)\le\left(\frac{2}{\xi}\right)^{n-s}(1+\xi\sqrt{n})^{n-2+2(n-s)}\int_Q\frac{1}{|x-z|^{n-2+(n-s)}}\,dx, \tag{5.7}$$

where the second integral is with respect to Lebesgue measure.

Proof. On the one hand, the Frostman-type bound $\mu(Q)\le(\operatorname{side}Q)^s$ and lower bound in (5.5) imply

$$\int_Q\frac{1}{|x-z|^{n-2}}\,d\mu(x)\le\frac{(\operatorname{side}Q)^n}{\operatorname{dist}(z,Q)^{n-2}(\operatorname{side}Q)^{n-s}}\le\left(\frac{2(1+\xi\sqrt{n})}{\xi}\right)^{n-s}\frac{(\operatorname{side}Q)^n}{\operatorname{dist}(z,Q)^{n-2+(n-s)}}.$$

On the other hand, if we write $\operatorname{ex}(Q,z)=\sup_{y\in Q}|y-z|$ for the *excess* of Q over z, then the upper bound in (5.5) implies $\operatorname{ex}(Q,z)\le\operatorname{dist}(z,Q)+\operatorname{diam}Q\le(1+\xi\sqrt{n})\operatorname{dist}(z,Q)$.

Hence

$$\int_Q \frac{1}{|x-z|^{n-2+(n-s)}}dx \geq \frac{(\operatorname{side} Q)^n}{\operatorname{ex}(Q,z)^{n-2+(n-s)}} \geq \frac{(\operatorname{side} Q)^n}{(1+\xi\sqrt{n})^{n-2+(n-s)}\operatorname{dist}(z,Q)^{n-2+(n-s)}}.$$

Merging the displayed equations yields (5.7). □

Theorem 5.9. *Let $n \geq 3$ and $h \geq n^3$ be integers. For all $s \in (n-1, n]$ satisfying $n - s \leq 0.2/(n^6 \log h)$,*

$$O(n, s; \Delta) \leq \left(1 + \frac{1}{n^6}\right) \frac{n\omega_n}{8}. \tag{5.8}$$

Remark. The upper bound in (5.8) depends on n and h, but otherwise is independent of the net Δ.

Proof. Let s satisfy the hypothesis. Then $(n-s)\log h \leq 0.2n^{-6} \leq 0.2/729 < \log 2$; in particular, $h^{n-s} \leq 2$. Fix $z \in \mathbb{R}^n$ and let μ be an admissible measure in the definition of $O_z(n, s; \Delta)$. We must bound $O_z(n, s; \Delta)$ from above independently of z.

We first dispense with the case when z is far away from the origin. Suppose that $|z| \geq 2$. Every cube $Q \in \Delta$ of side length h^{-1} that meets $B(0, 1/2)$ is contained in $B(0, (1/2) + \sqrt{n}h^{-1})$ and carries $\mu(Q) \leq h^{-s} = h^{n-s} \operatorname{vol} Q$. Since the cubes of side length h^{-1} in Δ are pairwise disjoint and μ is carried by $B^\Delta(0, 1/2)$, summing volumes gives

$$\mu(\mathbb{R}^n) \leq h^{n-s}\omega_n \left(\frac{1}{2} + \frac{\sqrt{n}}{h}\right)^n.$$

Because $|x - z| \geq |z| - (1/2) \geq 3/2$ for all $x \in B(0, 1/2)$, and because $h^{n-s} \leq 2$, $\sqrt{n}/h \leq n^{-2.5}$, and $(1 + 2n^{-2.5})^n \leq \exp(2n^{-1.5}) < 1.5$,

$$\int_{B^\Delta(0,1/2)} \frac{d\mu(x)}{|x-z|^{n-2}} \leq \frac{h^{n-s}\omega_n}{(3/2)^{n-2}} \left(\frac{1}{2} + \frac{\sqrt{n}}{h}\right)^n \leq \frac{n\omega_n}{8} \cdot \frac{54}{n\,3^n} \leq \frac{n\omega_n}{8}, \tag{5.9}$$

the last inequality because $n\,3^n \geq 81$ for all $n \geq 3$.

Suppose instead that $z \in B(0, 2)$. Fix $0 < \xi \leq (3h)^{-1}$ and let $\mathcal{W}$ be a family of cubes as in Lemma 5.7. Since μ is carried by $B^\Delta(0, 1/2)$ and the Frostman-type bound applied to cubes containing a point of $\mathsf{Site}(\Delta)$ at arbitrarily small scales gives zero mass to singletons, we have $\mu(\{z\}) = 0$. Thus, by Lemmas 5.7 and 5.8,

$$\begin{aligned}
\int_{B^\Delta(0,1/2)} \frac{1}{|x-z|^{n-2}}\, d\mu(x) &\leq \sum_{Q\in\mathcal{W}} \int_Q \frac{1}{|x-z|^{n-2}}\, d\mu(x) \\
&\leq \left(\frac{2}{\xi}\right)^{n-s} (1+\xi\sqrt{n})^{n-2+2(n-s)} \sum_{Q\in\mathcal{W}} \int_Q \frac{1}{|x-z|^{n-2+(n-s)}}\, dx \\
&\leq \left(\frac{2}{\xi}\right)^{n-s} (1+\xi\sqrt{n})^{n-2+2(n-s)} \int_{B(0,1/2+2.5\sqrt{n}\xi)} \frac{1}{|x-z|^{n-2+(n-s)}}\, dx.
\end{aligned} \tag{5.10}$$

For brevity, temporarily write $r = (1/2) + 2.5\sqrt{n}\xi$ and $\theta = n - 2 + (n-s)$. By Tonelli's theorem or layer-cake formula, and by monotonicity of volume,

$$\begin{aligned}\int_{B(0,r)} |x-z|^{-\theta}\,dx &= \int_0^\infty \mathcal{L}^n\left(B(z,t^{-1/\theta}) \cap B(0,r)\right) dt \\ &\le \int_0^\infty \min\left(\mathcal{L}^n(B(z,t^{-1/\theta})), \mathcal{L}^n(B(0,r))\right)\, dt \\ &= \int_0^\infty \mathcal{L}^n\left(B(0,t^{-1/\theta}) \cap B(0,r)\right) dt = \int_{B(0,r)} |x|^{-\theta}\,dx \\ &= n\omega_n \int_0^r t^{1-(n-s)}\,dt = \frac{n\omega_n}{2-(n-s)}\left(\frac{1}{2} + 2.5\sqrt{n}\xi\right)^{2-(n-s)}.\end{aligned} \tag{5.11}$$

Together (5.9), (5.10), and (5.11) imply

$$\begin{aligned}\sup_{z\in\mathbb{R}^n} O_z(n,s;\Delta) &\le \left(\frac{2}{\xi}\right)^{n-s} (1+\xi\sqrt{n})^{n-2+2(n-s)} \frac{n\omega_n}{2-(n-s)}\left(\frac{1}{2}+2.5\sqrt{n}\xi\right)^{2-(n-s)} \\ &= \frac{n\omega_n}{8}\left(\frac{4}{\xi}\right)^{n-s} (1+\xi\sqrt{n})^{n-2+2(n-s)} \frac{1}{1-\frac{n-s}{2}}\left(1+5\sqrt{n}\xi\right)^{2-(n-s)}.\end{aligned}$$

To derive the effective bound, set

$$\xi := \min\left(\frac{1}{3h}, \frac{1}{92n^{7.5}}\right) \tag{5.12}$$

and split the last display as $O(n,s;\Delta) \le (n\omega_n/8)\cdot A \cdot B$, where

$$A := (1+\xi\sqrt{n})^{n-2}\left(1+5\sqrt{n}\xi\right)^2, \qquad B := \frac{1}{1-\frac{n-s}{2}}\left(\frac{4(1+\xi\sqrt{n})^2}{\xi}\right)^{n-s}.$$

Since $(1+0.04n^{-6})(1+0.95n^{-6}) = 1 + 0.99n^{-6} + 0.038n^{-12} \le 1 + n^{-6}$, to prove (5.8) it suffices to bound A by $1+0.04n^{-6}$ and B by $1+0.95n^{-6}$.

From $1 + x \le e^x$ for all $x \in \mathbb{R}$ and $n + 8 \le 11n/3$, we obtain

$$A \le \exp\left(\sqrt{n}\xi(n+8)\right) \le \exp\left(\frac{11n^{1.5}\xi}{3}\right) \le \exp\left(\frac{11}{276n^6}\right).$$

Because $\log(1+x) \ge x - x^2/2$ and $n^{-6} \le 1/729$,

$$\log(1+0.04n^{-6}) \ge n^{-6}\left(0.04 - \frac{0.0008}{729}\right) > 0.0399n^{-6} > \frac{11}{276n^6}.$$

Therefore, $A \le 1 + 0.04n^{-6}$.

Next, $\sqrt{n}\xi \le (92n^7)^{-1} < 10^{-5}$, so that $4e^{0.51}(1+\xi\sqrt{n})^2 < 6.67$. Since $n - s \le 0.2/(n^6 \log h) < 10^{-4}$, we have $1/(1-\frac{n-s}{2}) \le e^{0.51(n-s)}$ and therefore

$$\log B \le (n-s)\log\frac{4e^{0.51}(1+\xi\sqrt{n})^2}{\xi} \le (n-s)\log\frac{6.67}{\xi}.$$

Since $h \geq n^3$ and $n \geq 3$, we have $\log n \leq \frac{1}{3}\log h$ and $\log h \geq 3\log 3 > 3.21$. Thus, by (5.12),

$$\begin{aligned}\log\frac{6.67}{\xi} &= \max\left(\log(20.01) + \log h,\ \log(613.64) + 7.5\log n\right)\\ &< \max\left(2.997 + \log h,\ 6.42 + 7.5\log n\right)\\ &\leq \max\left(2\log h,\ 2\log h + 2.5\log h\right) = 4.5\log h.\end{aligned}$$

Hence $\log B \leq 0.9n^{-6}$, while $\log(1+0.95n^{-6}) \geq 0.95n^{-6}(1-0.475n^{-6}) > 0.949n^{-6}$. Thus, $B \leq 1 + 0.95n^{-6}$, and we have proved (5.8). □

We now illustrate how to use the constants $O(n, s; \Delta)$ with other normalizations.

Lemma 5.10. *Let $n \geq 3$ and $h \geq 1$ be integers. Let $x_0 \in \mathbb{R}^n$, let $r > 0$, and let Δ be a dyadic net with initial scale $2rh^{-1}$. Let $s \in (n-1, n]$. If ν is a Borel measure with $\nu(\mathbb{R}^n \setminus B^\Delta(x_0, r)) = 0$ and $\nu(Q) \leq (\operatorname{side} Q)^s$ for all $Q \in \Delta$, then*

$$\sup_{z\in\mathbb{R}^n} \int_{B^\Delta(x_0,r)} \frac{1}{|x-z|^{n-2}}\, d\nu(x) \leq (2r)^{2-(n-s)} O(n, s; \widetilde{\Delta}),$$

where $\widetilde{\Delta}$ is the dyadic net with initial scale h^{-1} obtained by applying the homeomorphism $y \mapsto f(y) = (y - x_0)/(2r)$ to Δ.

Proof. Note that f maps $\mathsf{Site}(\Delta)$ onto $\mathsf{Site}(\widetilde{\Delta})$ and $B(x_0, r)$ onto $B(0, 1/2)$; hence f maps $B^\Delta(x_0, r)$ onto $B^{\widetilde{\Delta}}(0, 1/2)$. Define the image measure $\mu = (2r)^{-s} f_\sharp \nu$ by the rule $\mu(E) = (2r)^{-s}\nu(f^{-1}(E))$ for all Borel sets $E \subset \mathbb{R}^n$. In particular, $\mu(\mathbb{R}^n \setminus B^{\widetilde{\Delta}}(0, 1/2)) = 0$ and

$$\mu(Q) = (2r)^{-s}\nu(x_0 + 2rQ) \leq (2r)^{-s}(\operatorname{side}(x_0 + 2rQ))^s = (\operatorname{side} Q)^s$$

for all $Q \in \widetilde{\Delta}$. Hence μ is an admissible measure for the constant $O(n, s; \widetilde{\Delta})$. Let $w \in \mathbb{R}^n$, let $z = f(w)$, and define $g(x) = |x - z|^{-(n-2)}$ for all $x \in \mathbb{R}^n$. By the change-of-variables formula $\int g\, df_\sharp\nu = \int g \circ f\, d\nu$ for image measures, e.g., see [Mat95, Theorem 1.19], we have

$$\begin{aligned}O(n, s; \widetilde{\Delta}) \geq \int_{B^{\widetilde{\Delta}}(0,1/2)} \frac{1}{|x-z|^{n-2}}\, d\mu(x) &= \frac{1}{(2r)^s}\int_{B^\Delta(x_0,r)} \frac{1}{|f(y) - f(w)|^{n-2}}\, d\nu(y)\\ &= \frac{(2r)^{n-2}}{(2r)^s}\int_{B^\Delta(x_0,r)} \frac{1}{|y-w|^{n-2}}\, d\nu(y).\end{aligned}$$

Rearranging and taking the supremum over $w \in \mathbb{R}^n$ completes the proof. □

By an abuse of notation, given integers $n \geq 3$ and $h \geq 1$, a point $x_0 \in \mathbb{R}^n$, a number $r > 0$, a number $s \in (n-1, n]$, and a dyadic net Δ with initial scale $2rh^{-1}$, we write $O(n, s; \Delta)$ for $O(n, s; \widetilde{\Delta})$, where $\widetilde{\Delta}$ is the net at initial scale h^{-1} obtained from Δ as in Lemma 5.10. The value so defined depends on the normalization, but our estimates use this constant only through upper bounds that are uniform over nets at the normalized scale. In particular, when $h \geq n^3$ and $n - s \leq 0.2/(n^6 \log h)$, Theorem 5.9 shows that (5.8) holds independently of the particular normalized net $\widetilde{\Delta}$. Thus, under these assumptions, (5.8) applies to $O(n, s; \Delta)$ under any normalization. In Lemma 5.11 below, the data h, x_0, and r are supplied by the hypotheses.

5.3. **Bourgain's estimate, again.** The bounds on $O(n, s; \Delta)$ in Theorem 5.9 yield an improved effective version of Bourgain's estimate for harmonic measure; compare to [Bou87, Lemma 1] and [BG24, Lemma 3.1]. Here and below, if $V \subset \mathbb{R}^n$ is a bounded open set and $x \in V$, then ω_V^x denotes the exit distribution of Brownian motion started at x and run until it first leaves V; equivalently, $\omega_V^x = \omega_{V^x}^x$, where V^x is the connected component of V containing x. Note that V^x is open and $\partial V^x \subseteq \partial V$. For nonempty sets $A, B \subset \mathbb{R}^n$, we write $\operatorname{gap}(A, B) := \inf_{a\in A} \inf_{b\in B} |a - b|$ for the *gap* between A and B (often referred to as the *distance* between A and B, despite failing the triangle inequality).

Lemma 5.11 (Bourgain's estimate for a round snapshot with a nearly sharp constant)**.** *Let $n \geq 3$ and $h \geq n^3$ be integers and let $U \subset \mathbb{R}^n$ be a bounded domain. Suppose that the open ball $B(x_0, r) \subset U$ satisfies $\operatorname{gap}(B(x_0, r), \partial U) = gr$ for some $g > 1$. Let Δ be any dyadic net with initial scale $2r \cdot h^{-1}$ such that $\mathsf{Site}(\Delta) \cap \overline{B(x_0, r)}$ is compact. For all closed $E \subset \mathbb{R}^n$ with $x_0 \notin E$ and for all $s \leq n$ satisfying $s \geq n - 0.2/(n^6 \log h)$,*

$$\omega_{U\setminus E}^{x_0}(E \cap U) \geq \frac{2}{n} \cdot \frac{1 - g^{-(n-2)}}{1 + n^{-6}} \cdot \frac{\mathcal{M}_\infty^{\Delta,s}(E \cap B(x_0, r))}{\omega_n r^s}. \tag{5.13}$$

Proof. Let $E \subset \mathbb{R}^n$ be closed. We may assume that $\mathcal{M}_\infty^{\Delta,s}(E \cap B(x_0, r)) > 0$, since otherwise the right hand side of (5.13) is zero and the estimate is trivial. Then, under the assumption that $\mathsf{Site}(\Delta) \cap \overline{B(x_0, r)}$ is compact, the set $K := E \cap \overline{B(x_0, r)} \cap \mathsf{Site}(\Delta)$ is compact and $K \subset \mathsf{Site}(\Delta)$. Let s satisfy the bounds in the hypothesis and note that $s \in (n - 1, n]$, since $0.2/(n^6 \log h) < 1$ for all $n \geq 3$ and $h \geq n^3$. Note that $\partial B(x_0, r)$ has $\mathcal{M}_\infty^{\Delta,s}$ measure zero, since $\partial B(x_0, r)$ has Hausdorff dimension $n - 1$ and $s > n - 1$. Hence $\mathcal{M}_\infty^{\Delta,s}(K) = \mathcal{M}_\infty^{\Delta,s}(E \cap B(x_0, r))$. By Lemma 5.4, there exists a finite Borel measure μ on $\mathbb{R}^n$ with support in K such that $\mu(Q) \leq (\operatorname{side} Q)^s$ for all $Q \in \Delta$ and $\mu(\mathbb{R}^n) = \mu(K) \geq \mathcal{M}_\infty^{\Delta,s}(K)$. The measure μ does not charge $\partial B(x_0, r)$. Indeed, given $\epsilon > 0$, pick cubes $Q_1, Q_2, \cdots \in \Delta$ that cover $\partial B(x_0, r) \cap \mathsf{Site}(\Delta)$ with $\sum_i (\operatorname{side} Q_i)^s < \epsilon$; since μ is carried by $K \subset \mathsf{Site}(\Delta)$,

$$\mu(\partial B(x_0, r)) \leq \sum_i \mu(Q_i) \leq \sum_i (\operatorname{side} Q_i)^s < \epsilon.$$

Thus, $\mu(\mathbb{R}^n \setminus B^\Delta(x_0, r)) = 0$, so that μ is admissible in Lemma 5.10.

Consider the harmonic function $u(z) = \int_K |x - z|^{-(n-2)}\, d\mu(x)$ defined on $\mathbb{R}^n \setminus K$. By Lemma 5.10, the inequality $2^{2-(n-s)} \leq 4$, and elementary estimates, we have

$$u(z) \leq (2r)^{2-(n-s)} O(n, s; \Delta) \leq 4 \cdot r^{2-(n-s)} O(n, s; \Delta) \quad \forall\, z \in \mathbb{R}^n \setminus K, \tag{5.14}$$

$$u(x_0) \geq r^{-(n-2)} \mu(K), \tag{5.15}$$

$$u(x) \leq (\operatorname{gap}(B(x_0, r), \partial U))^{-(n-2)} \mu(K) = (gr)^{-(n-2)} \mu(K) \quad \forall\, x \in \partial U. \tag{5.16}$$

The estimates (5.15) and (5.16) are straightforward. Since $\mu(K) \geq \mathcal{M}_\infty^{\Delta,s}(K) > 0$, we have $u > 0$ on $\mathbb{R}^n \setminus K$; in particular, $\|u\|_{L^\infty(\mathbb{R}^n \setminus K)} \in (0, \infty)$ by (5.14). Following [Bou87],

define an auxiliary harmonic function w on $\mathbb{R}^n \setminus K$ by setting

$$w(x) = \frac{u(x) - \|u\|_{L^\infty(\partial U)}}{\|u\|_{L^\infty(\mathbb{R}^n \setminus K)}} \quad \text{for all } x \in \mathbb{R}^n \setminus K,$$

where the norms denote the supremum of the continuous function u on the specified sets. By design, $w \le 0$ on ∂U, w is continuous on ∂U, and $w \le 1$ on all of $\mathbb{R}^n \setminus K$. Hence

$$\limsup_{y \to x} w(y) \le \chi_K(x) \quad \text{for all } x \in \partial(U \setminus K).$$

Thus, $w(x) \le \omega^x_{U \setminus K}(K)$ for all $x \in U \setminus K$ by the maximum principle for harmonic functions.

We next compare this harmonic measure with the one in the statement of the lemma. Since $U \setminus E \subseteq U \setminus K$ and $K \subseteq E$, any Brownian path that reaches K before exiting U must first leave $U \setminus E$ through $E \cap U$. Therefore,

$$\omega^x_{U \setminus K}(K) \le \omega^x_{U \setminus E}(E \cap U) \quad \text{for all } x \in U \setminus E.$$

Using (5.14) to estimate the denominator in the definition of w and (5.15), (5.16) to estimate the numerator, we obtain

$$\omega^{x_0}_{U \setminus E}(U \cap E) \ge \frac{r^{-(n-2)}(1 - g^{-(n-2)})\mathcal{M}^{\Delta,s}_\infty(E \cap B(x_0, r))}{4 \cdot r^{2-(n-s)} O(n, s; \Delta)}.$$

Applying Theorem 5.9 to bound $O(n, s; \Delta)$ and rearranging yields (5.13). □

Example 5.12. The phrase "nearly sharp constant" in the title of Lemma 5.11 refers to a constant obtained by following the skeleton of Bourgain's original proof rather than to an assertion that the constant in (5.13) is optimal among all possible proofs. For example, take $x_0 = 0$, $r = 1/2$, and $s = n$, and suppose that Δ has initial scale 1 with $\mathsf{Site}(\Delta) \supset \overline{B(0, 1/2)}$. For $0 < \eta \ll 1/2$, let

$$C_\eta = \{(x_1, \dots, x_n) \in \mathbb{R}^n : x_1 > -\eta \text{ and } x_2^2 + \cdots + x_n^2 < \eta^2\}, \qquad K = \overline{B(0, 1/2)} \setminus C_\eta,$$

and let μ be Lebesgue measure restricted to K. Then μ is an n-dimensional Frostman measure for K relative to Δ, with $\mu(K) = \mathcal{M}^{\Delta,n}_\infty(K)$. Thus, we can use μ in Bourgain's argument. If $u(z) = \int_K |x - z|^{-(n-2)}\, d\mu(x)$, then

$$\|u\|_{L^\infty(\mathbb{R}^n \setminus K)} \ge u(0) = \int_{B(0,1/2) \setminus C_\eta} \frac{dx}{|x|^{n-2}} \longrightarrow \frac{n\omega_n}{8} \quad \text{as } \eta \downarrow 0.$$

On the other hand, Theorem 5.9 gives $\|u\|_{L^\infty(\mathbb{R}^n \setminus K)} \le O(n, n; \Delta) \le (1 + n^{-6}) n\omega_n/8$. Hence the estimate for the Frostman potential used in the proof is nearly attained in this example. A different proof of Lemma 5.11 might conceivably yield a better constant.

6. One-ring alternative

Recall from the introduction that d_n is the supremum of the upper Hausdorff dimensions of harmonic measure over bounded domains in $\mathbb{R}^n$. We follow Bourgain's original strategy for bounding d_n [Bou87], as updated in [BG24], but introduce two new estimates that substantially strengthen the barrier argument. Lemma 6.6 exploits repeated returns of

Brownian motion to a separating surface, raising the lower bound on the probability of hitting the boundary inside an annular ring. Lemma 6.7 then allows us to use just one "thin ring without holes" in place of Bourgain's original multi-ring barrier. We also use the covers by cubes in general position and the Bourgain estimate for round snapshots with a nearly sharp constant developed above.

6.1. **Overview.** Fix $n \geq 3$. The model depends on the following parameters:

- integers $M \geq 3$, $L \geq 2$, with $2L \geq 3M$, and $h \geq n^3$, which control the geometry and scales of the cube partitions described in §6.2;
- $0 < \rho \leq \rho_0$ and $0 < a < 1$ that determine the size of an $(n-\rho)$-dimensional hole, where the upper bound on ρ is

$$\rho_0 := \frac{0.2}{n^6 \log h}; \tag{6.1}$$

  see (6.7);
- a lower bound η for harmonic measure in the absence of a hole, given by

$$\eta := \frac{2a}{n + n^{-5}}(1 - 2^{-(n-2)}); \tag{6.2}$$

  see Lemma 6.3.

The following theorem is the main result of this section. It converts any admissible set of parameters into an explicit upper bound for d_n.

Theorem 6.1. *Let $n \geq 3$ and let $0 < a < 1$. Let $M \geq 3$, $L \geq 2$, and $h \geq n^3$ be integers with $2L \geq 3M$ such that*

$$W := \left(1 + \frac{n\,\omega_n}{\omega_{n-1}} \left(\frac{4L}{3M} - 1\right) \eta\right)^{-1} < \left(1 - \frac{2}{M}\right)^n =: V, \tag{6.3}$$

and let $\rho \in (0, \rho_0]$ satisfy

$$\begin{aligned} \left(1 - \frac{1}{L}\right)^{n-\rho} + \left(L^n - (L-1)^n - 2^{n-1}\right) \left(\frac{1}{L}\right)^{n-\rho} + 2^{n-1}(2^n - 1) \left(\frac{1}{2L}\right)^{n-\rho} \\ + \left(2^{n-1} h^n - \frac{\omega_n}{2^n} \left(h - 2\sqrt{n}\right)^n\right) \left(\frac{1}{2Lh}\right)^{n-\rho} + a\omega_n \left(\frac{1}{4L}\right)^{n-\rho} \leq 1. \end{aligned} \tag{6.4}$$

Then $d_n \leq n - \lambda\rho/(1+\lambda)$, where $\lambda > 0$ is given by

$$\lambda := W \log_\tau \frac{W}{V} + (1 - W) \log_\tau \frac{1 - W}{1 - V}, \tag{6.5}$$

$$\tau := \left(1 - \frac{2}{M}\right)^{n-\rho} + \left(M^n - (M-2)^n\right) \left(\frac{1}{M}\right)^{n-\rho} \qquad (\tau > 1). \tag{6.6}$$

After describing the one-ring alternative in §6.2, we establish lemmata in §6.3 and record the proof of Theorem 6.1 in §6.4. Suitable choices of parameters are then used to prove Theorem 1.1 in §7.

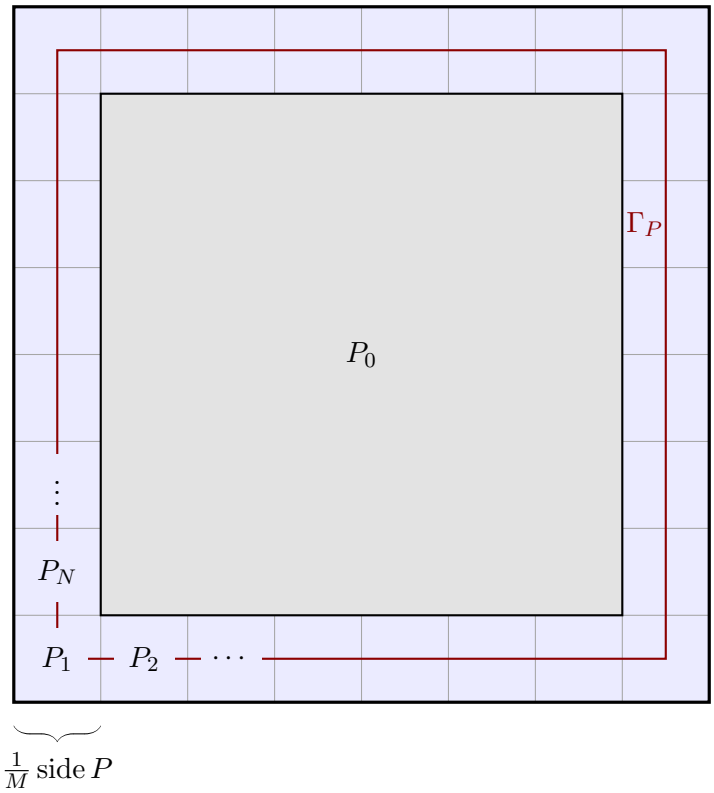


FIGURE 6.1. Illustrated with $n = 2$, $M = 8$, and $L = 12$: the partition used in **Alternative II**. The outer children form the ring R_P around the inner child P_0, and the separating surface Γ_P lies inside the ring; for the parameters shown, it lies midway across R_P.

6.2. **One-ring alternative.** Let $\Omega \subset \mathbb{R}^n$ be a bounded domain and set $E := \partial\Omega$. Thus, E is closed and harmonic measure on Ω is supported on E. Fix a cube P in $\mathbb{R}^n$ of any type. We examine the geometry of E inside P in order to show that P satisfies condition (I) or (II) of Theorem 4.1. The model uses two related partitions of P, one for each alternative. In **Alternative I**, the partition is adapted to the presence of a hole in E. In **Alternative II**, the partition is used when no such hole is present.

6.2.1. *First partition: the annular ring.* Fix integers $M \geq 3$ and $L \geq 2$ with $2L \geq 3M$ (independent of P). Partition P into subcubes (see Figure 6.1): an *inner child* P_0 of side length $(1 - 2/M)$ side P and $N = M^n - (M-2)^n$ *outer children* $P_1, \ldots, P_N$ of side length $(1/M)$ side P, arranged so that P_0 contains the center of P and is surrounded by $P_1, \ldots, P_N$. As usual, the children may have different types; see the paragraph above Definition 3.1. The outer children form a *ring* $R_P = \bigcup_{i=1}^{N} P_i$ of width $(1/M)$ side P around the inner child.

Define the *separating surface* Γ_P to be the boundary of any concentric subcube of P of side length $(1 - 3/2L)$ side P. Thus, Γ_P lies at distance $(3/4L)$ side P from ∂P and at distance $(1/M - 3/4L)$ side P from ∂P_0. Because $2L \geq 3M$, the first distance is at most

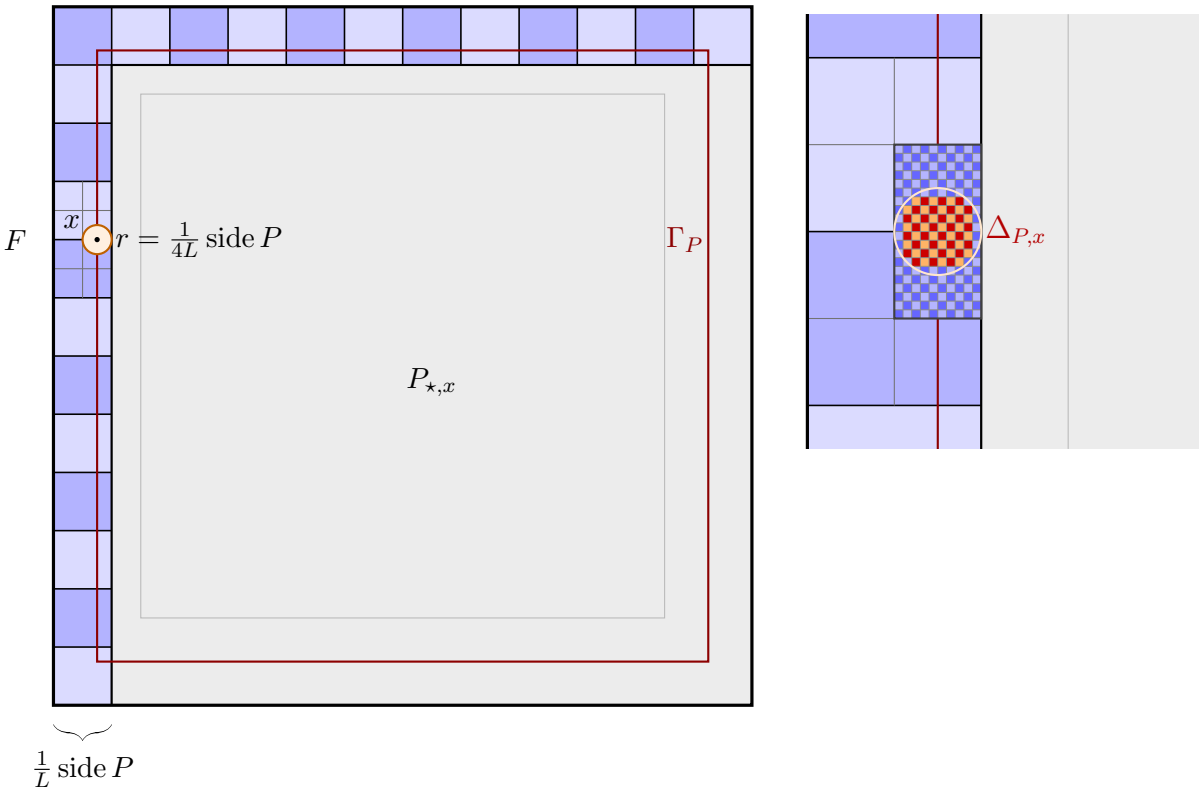


FIGURE 6.2. Illustrated with $n = 2$, $M = 8$, $L = 12$, and $h = 10$: the x-dependent partition used in **Alternative I**. The 'L'-shaped collar is divided into medium cubes; only those near $B_{P,x}$ are subdivided into little cubes, and only the little cubes covering $B_{P,x}$ are refined into tiny cubes. The tiny cubes contained in $B_{P,x}$ are the top cubes of $\Delta_{P,x}$.

the second. Hence Γ_P lies inside R_P, at least as close to ∂P as to ∂P_0. For $x \in \Gamma_P$, set

$$B_{P,x} := B(x, (1/4L)\,\text{side}\,P), \qquad \widehat{B}_{P,x} := B(x, (3/4L)\,\text{side}\,P).$$

The two balls are concentric and $B_{P,x} \subset \widehat{B}_{P,x} \subset R_P$.

6.2.2. *Second partition: the L-shaped collar.* Fix $x \in \Gamma_P$. To define a hole at x, we form a second partition of P that depends on x. See Figure 6.2. It is not too important to remember the names of the various classes of cubes below; the essential information is the size and number of each type, which can be looked up as needed.

Choose a face F of P at distance $(3/4L)\,\text{side}\,P$ from x and choose any *big cube* $P_{\star,x} \subset P$ of side length $(1-1/L)\,\text{side}\,P$ so that half of the faces of $P_{\star,x}$ are contained in the faces of P (i.e. $P_{\star,x}$ and P share a corner) and so that the face of $P_{\star,x}$ parallel to F lies at distance $(1/L)\,\text{side}\,P$ from F. In particular, $x \notin P_{\star,x}$.

Next, partition the 'L'-shaped *collar* $P \setminus P_{\star,x}$ into $L^n - (L-1)^n$ *medium cubes* of side length $(1/L)\,\text{side}\,P$. Since x lies at distance $(3/4L)\,\text{side}\,P$ from F, the points of $B_{P,x}$ lie at distances between $(1/2L)\,\text{side}\,P$ and $(1/L)\,\text{side}\,P$ from F. Thus, $B_{P,x}$ is contained in

the union of the medium cubes whose closures meet F. These cubes vary only in the $n-1$ coordinate directions parallel to F. Since the diameter of $B_{P,x}$ is half the side length of a medium cube, it can meet at most two such cubes in each of these directions, and hence at most 2^{n-1} medium cubes in total. Consequently, we may choose a family $\mathcal{M}_{P,x}$ of exactly 2^{n-1} medium cubes whose union contains $B_{P,x}$.

Partition each cube in $\mathcal{M}_{P,x}$ into 2^n *little cubes* of side length $(1/2L)\operatorname{side} P$. Since the points of $B_{P,x}$ lie at distances between $(1/2L)\operatorname{side} P$ and $(1/L)\operatorname{side} P$ from F, the ball is contained in the union of the little cubes farthest from F within the cubes of $\mathcal{M}_{P,x}$. These little cubes vary only in the $n-1$ coordinate directions parallel to F. Since the diameter of $B_{P,x}$ equals the side length of a little cube, it can meet at most two such cubes in each of these directions, and hence at most 2^{n-1} little cubes in total. Choose a family $\mathcal{L}_{P,x}$ of exactly 2^{n-1} little cubes whose union contains $B_{P,x}$. The cubes in $\mathcal{L}_{P,x}$ need not belong to the same medium cube.

We now refine the little cubes in $\mathcal{L}_{P,x}$. Let $h \geq n^3$ be an integer independent of P and x, and partition each cube in $\mathcal{L}_{P,x}$ into *tiny cubes* of side length $h^{-1}\operatorname{diam} B_{P,x} = (1/2Lh)\operatorname{side} P$. Let $\mathcal{T}_{P,x}$ denote the family of tiny cubes contained in $B_{P,x}$. Because $\overline{B_{P,x}} \subset \operatorname{int} P$, we may reassign boundaries of the big, medium, little, and tiny cubes without altering the boundary of P so that $\bigcup \mathcal{T}_{P,x}$ is compact. Choose any dyadic net $\Delta_{P,x}$ with initial scale $(1/2Lh)\operatorname{side} P$ whose top cubes are $\mathcal{T}_{P,x}$; see Definition 5.1. For emphasis, we note that $\mathsf{Site}(\Delta_{P,x}) = \bigcup \mathcal{T}_{P,x}$ is compact.

6.2.3. *Holes.* Let $0 < a < 1$ be a model parameter independent of P and x. We say that there is a *hole* in P at $x \in \Gamma_P$ if

$$\mathcal{M}_\infty^{\Delta, n-\rho}(E) = \mathcal{M}_\infty^{\Delta, n-\rho}(\partial\Omega \cap \bigcup \mathcal{T}_{P,x}) < a \cdot \omega_n\big((1/4L)\operatorname{side} P\big)^{n-\rho}, \tag{6.7}$$

where $\Delta = \Delta_{P,x}$.

We say that P satisfies **Alternative I** if there is a hole in P at some $x \in \Gamma_P$. The existence of such a hole will allow us to construct an efficient $(n-\rho)$-dimensional cover of $E \cap P$. Otherwise, P satisfies **Alternative II**: there is no hole in P at any $x \in \Gamma_P$. The absence of holes gives a uniformly positive probability of hitting E locally along the separating surface and ultimately limits how much harmonic measure can reach the inner child P_0. Every cube P satisfies exactly one of the two alternatives.

6.3. **Lemmata.**

Lemma 6.2. *Suppose that P satisfies **Alternative I**. There is a covering of $E \cap P$ by a set $\mathcal{Q}_P$ of pairwise disjoint cubes with* $\operatorname{side} Q \leq (1-1/L)\operatorname{side} P$ *for all $Q \in \mathcal{Q}_P$ and*

$$\sum_{Q \in \mathcal{Q}_P} (\operatorname{side} Q)^{n-\rho} \leq (\textit{left hand side of } (6.4)) \cdot (\operatorname{side} P)^{n-\rho}.$$

Proof. Without loss of generality, we may assume that $\operatorname{side} P = 1$. Because P satisfies **Alternative I**, there exists a hole in P at some $x \in \Gamma_P$. Consider the associated partition of P described in Figure 6.2. In the collection $\mathcal{Q}_P$, we include the big cube $P_{\star,x}$, the medium cubes in $P \setminus P_{\star,x}$ that do not belong to $\mathcal{M}_{P,x}$, the little cubes contained in

$\bigcup \mathcal{M}_{P,x} \setminus \bigcup \mathcal{L}_{P,x}$, and the tiny cubes in $\mathcal{L}_{P,x}$ that are not contained in $B_{P,x}$. There are $L^n - (L-1)^n - 2^{n-1}$ such medium cubes and $2^{n-1}(2^n - 1)$ such little cubes. Thus, the big cube, the medium cubes, and the little cubes in $\mathcal{Q}_P$ contribute the first three terms in (6.4) to its $(n-\rho)$-dimensional cubical content. Now, if a tiny cube T is not contained in $B_{P,x}$, then T is contained in $\bigcup \mathcal{L}_{P,x} \setminus B(x, 1/(4L) - \sqrt{n}/(2Lh))$. Dividing the volume of this set by the volume of a tiny cube shows that the number of such tiny cubes is at most $2^{n-1}h^n - (\omega_n/2^n)(h - 2\sqrt{n})^n$. Hence these tiny cubes contribute at most the fourth term in (6.4) to the $(n-\rho)$-dimensional cubical content of $\mathcal{Q}_P$. Finally, since $\mathcal{M}_\infty^{\Delta, n-\rho}(E)$ is strictly less than the fifth term in (6.4) by the definition of a hole (6.7), Lemma 5.3 applied with $s = n - \rho$ furnishes countably many pairwise disjoint cubes contained in $\mathsf{Site}(\Delta_{P,x}) = \bigcup \mathcal{T}_{P,x}$ that cover $E \cap \mathsf{Site}(\Delta_{P,x})$ and whose total $(n-\rho)$-dimensional cubical content is at most the fifth term; these are the final cubes that we include in $\mathcal{Q}_P$. The collection $\mathcal{Q}_P$ has the desired properties: its cubes are pairwise disjoint, have side length no greater than that of the big cube, cover $E \cap P$, and satisfy the asserted estimate. □

Lemma 6.3 (locally large harmonic measure)**.** *Suppose that P satisfies **Alternative II**. For all $x \in \Gamma_P \cap \Omega$, we have $\widehat{B}_{P,x} \subset R_P$ and*

$$\omega^x_{\widehat{B}_{P,x} \setminus E}(E \cap \widehat{B}_{P,x}) \geq \eta. \tag{6.8}$$

Proof. Let $x \in \Gamma_P \cap \Omega$. The inclusion $\widehat{B}_{P,x} \subset R_P$ was noted in §6.2. Assign $s = n - \rho$, $U = \widehat{B}_{P,x}$, $r = (1/4L)\operatorname{side} P$, $g = 2$, and $\Delta = \Delta_{P,x}$. Then $B(x,r) = B_{P,x}$ and the hypothesis of Lemma 5.11 is satisfied. Indeed, s is in the valid range of exponents, because $\rho \leq \rho_0$. We also have $\operatorname{gap}(B_{P,x}, \partial U) = (3/4L)\operatorname{side} P - (1/4L)\operatorname{side} P = 2r = gr$. Finally, the cubes in Δ have initial scale $2r/h$, and $\mathsf{Site}(\Delta_{P,x}) \cap \overline{B_{P,x}}$ is compact, because $\mathsf{Site}(\Delta_{P,x})$ is compact. Now, the opposite inequality to (6.7) holds, because P satisfies **Alternative II**. Inserting it into (5.13), we obtain

$$\omega^x_{U \setminus E}(E \cap U) \geq \frac{2}{n} \cdot \frac{1 - 2^{-(n-2)}}{1 + n^{-6}} \cdot \frac{\mathcal{M}_\infty^{\Delta, n-\rho}(E \cap B(x,r))}{\omega_n r^{n-\rho}} \geq \frac{2}{n} \cdot \frac{1 - 2^{-(n-2)}}{1 + n^{-6}} \cdot \frac{a\omega_n r^{n-\rho}}{\omega_n r^{n-\rho}} = \eta,$$

with η defined as in (6.2). □

Lemma 6.4 (strong Markov property for harmonic measure)**.** *Let $\Omega_1 \subset \Omega_2 \subset \mathbb{R}^n$ be bounded domains. Then for every $x \in \Omega_1$ and every Borel set $A \subset \partial\Omega_2$,*

$$\omega^x_{\Omega_2}(A) \;=\; \omega^x_{\Omega_1}\big(A \cap \partial\Omega_1\big) \;+\; \int_{\partial\Omega_1 \cap \Omega_2} \omega^z_{\Omega_2}(A)\, d\omega^x_{\Omega_1}(z). \tag{6.9}$$

We will use the following immediate consequences. If $A, B \subset \partial\Omega_2$ are Borel sets and $c \in \mathbb{R}$, write $\varphi(y) := \omega^y_{\Omega_2}(A) - c\,\omega^y_{\Omega_2}(B)$ for $y \in \Omega_2$. Then, for every $x \in \Omega_1$,

$$\omega^x_{\Omega_1}(A \cap \partial\Omega_1) \leq \omega^x_{\Omega_2}(A) \quad \textit{and} \quad \int_{\partial\Omega_1 \cap \Omega_2} \omega^z_{\Omega_2}(A)\, d\omega^x_{\Omega_1}(z) \leq \omega^x_{\Omega_2}(A), \tag{6.9a}$$

$$\omega^x_{\Omega_2}(A) = \int_{\partial\Omega_1 \cap \Omega_2} \omega^z_{\Omega_2}(A)\, d\omega^x_{\Omega_1}(z) \quad \textit{if } \omega^x_{\Omega_1}(A \cap \partial\Omega_1) = 0, \tag{6.9b}$$

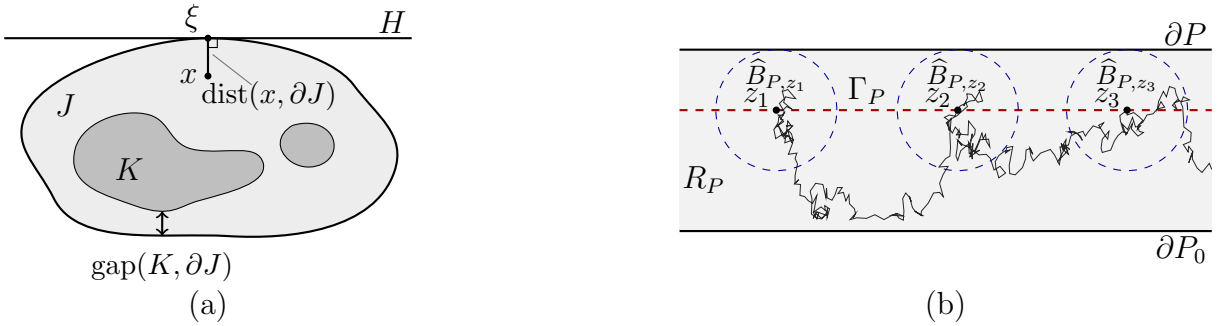


FIGURE 6.3. (a) Lemma 6.5: a supporting hyperplane H for J at a point $\xi \in \partial J$ nearest to $x \in J \setminus K$. (b) Lemma 6.6: a Brownian path departing the separating surface from $z_1 \in \Gamma_P$ may return to Γ_P at $z_2, z_3, \ldots$ without leaving the ring R_P, where z_{i+1} is the first point of Γ_P met after the path leaves $\widehat{B}_{P,z_i}$. Under **Alternative II**, at each z_i there is uniformly large probability that the path meets $\partial\Omega$ inside $\widehat{B}_{P,z_i}$.

$$\varphi(x) = \omega^x_{\Omega_1}(A \cap \partial\Omega_1) - c\,\omega^x_{\Omega_1}(B \cap \partial\Omega_1) + \int_{\partial\Omega_1 \cap \Omega_2} \varphi \, d\omega^x_{\Omega_1}, \tag{6.9c}$$

$$\varphi(x) \le \omega^x_{\Omega_1}(A \cap \partial\Omega_1) + \int_{\partial\Omega_1 \cap \Omega_2} \varphi \, d\omega^x_{\Omega_1} \quad \text{if } c \ge 0. \tag{6.9d}$$

Proof. For (6.9), see e.g. [AAM19, Lemma B.1] or [BG25, Theorem 3.8]. Since the two terms on the right hand side are nonnegative, (6.9a) follows, and (6.9b) is the case in which the first term vanishes. Applying (6.9) separately to A and B and subtracting c times the second identity gives (6.9c). If $c \ge 0$, omitting the nonpositive term $-c\,\omega^x_{\Omega_1}(B \cap \partial\Omega_1)$ gives (6.9d). □

Lemma 6.5. *Let $J \subset \mathbb{R}^n$ be a bounded open convex set and let $K \subset J$ be compact. If U is a domain contained in $J \setminus K$, then*

$$\omega^x_U(\partial K) \le \frac{\operatorname{dist}(x, \partial J)}{\operatorname{gap}(K, \partial J)} \qquad \text{for all } x \in U. \tag{6.10}$$

Proof. Let S be the connected component of $J \setminus K$ containing U. Fix $x \in U$ and choose $\xi \in \partial J$ with $|x - \xi| = \operatorname{dist}(x, \partial J)$. If H is a hyperplane supporting J at ξ, let ℓ be the affine function with $\ell(y) = \operatorname{dist}(y, H)$ for $y \in J$. Then $\ell(x) \le \operatorname{dist}(x, \partial J)$. For $y \in K$, the segment from y to the nearest point of H ends outside J and therefore meets ∂J, so $\ell(y) \ge \operatorname{gap}(K, \partial J)$. Since $\ell \ge 0$ on $\overline{J}$ and ℓ is harmonic,

$$\operatorname{dist}(x, \partial J) \ge \ell(x) = \int_{\partial S} \ell \, d\omega^x_S \ge \operatorname{gap}(K, \partial J)\, \omega^x_S(\partial K).$$

Moreover, $\partial K \cap \partial U \subset \partial K \cap \partial S$. The first inequality in (6.9a), applied to $U \subset S$ with $A = \partial K \cap \partial S$, gives $\omega^x_U(\partial K) \le \omega^x_S(\partial K)$. □

Figure 6.3(b) suggests the probabilistic idea behind the next lemma. Under **Alternative II**, at each return to Γ_P at z, the path has probability at least η of hitting E inside $\widehat{B}_{P,z} \subset R_P$. Carrying out this idea gives the constant θ below. The proof is self-contained, uses only harmonic measure, and encodes all returns in one supremum.

Lemma 6.6. *Suppose that P satisfies **Alternative II**. If $x_0 \in \Omega \setminus P$ and $\omega \equiv \omega_\Omega^{x_0}$, then*

$$\omega(P_0) \le \theta\, \omega(R_P), \qquad \theta := \left(\frac{n\,\omega_n}{\omega_{n-1}} \left(\frac{4L}{3M} - 1 \right) \eta \right)^{-1}. \tag{6.11}$$

Proof. Let S be the closed cube with $\partial S = \Gamma_P$. Set

$$r := \frac{3}{4L} \operatorname{side} P, \qquad d := \left(\frac{1}{M} - \frac{3}{4L} \right) \operatorname{side} P, \qquad \widehat{B}_{P,z} = B(z,r), \qquad \operatorname{gap}(\overline{P_0}, \Gamma_P) = d.$$

Then $r \le d$ by $2L \ge 3M$, and $\theta\eta = (r/d)\omega_{n-1}/(n\omega_n)$. For $y \in \Omega$, set

$$\varphi(y) := \omega_\Omega^y(P_0) - \theta\, \omega_\Omega^y(R_P).$$

Since harmonic measure is supported on E, φ is the function in (6.9c) and (6.9d) with $A = E \cap P_0$, $B = E \cap R_P$, and $c = \theta$. In particular, $\varphi \le 1$. If $\Gamma_P \cap \Omega = \emptyset$, connectedness and $x_0 \notin S$ force $\Omega \cap S = \emptyset$, so $\omega(P_0) = 0$. Thus, assume that $\Gamma_P \cap \Omega \ne \emptyset$ and write $\Gamma := \Gamma_P \cap \Omega$. It suffices to prove $\sup_\Gamma \varphi \le 0$. Indeed, let Ω' be the component of $\Omega \setminus S$ containing x_0. Then $\partial\Omega' \cap \Omega \subset \Gamma$ and $(E \cap P_0) \cap \partial\Omega' = \emptyset$. If $\sup_\Gamma \varphi \le 0$, then

$$\omega_\Omega^y(P_0) \le \theta\, \omega_\Omega^y(R_P) \qquad \text{for all } y \in \partial\Omega' \cap \Omega.$$

Hence (6.9b) and (6.9a) give

$$\omega(P_0) = \int_{\partial\Omega' \cap \Omega} \omega_\Omega^y(P_0)\, d\omega_{\Omega'}^{x_0}(y) \le \theta \int_{\partial\Omega' \cap \Omega} \omega_\Omega^y(R_P)\, d\omega_{\Omega'}^{x_0}(y) \le \theta\, \omega(R_P).$$

Set $C = \max\{0, \sup_\Gamma \varphi\}$. Since $\varphi \le 1$, we have $0 \le C \le 1$. To complete the proof, it remains to show that $C = 0$.

First let $y \in \Omega \setminus S$, and let U be the component of $\Omega \setminus S$ containing y. Then $\partial U \cap \Omega \subset \Gamma$ and $\partial U \cap \overline{P_0} = \emptyset$. By (6.9c),

$$\varphi(y) = -\theta\, \omega_U^y((E \cap R_P) \cap \partial U) + \int_{\partial U \cap \Omega} \varphi\, d\omega_U^y \le C.$$

Next let $y \in \Omega \cap \operatorname{int} S \setminus \overline{P_0}$, and let V be the component of $\Omega \cap \operatorname{int} S \setminus \overline{P_0}$ containing y. Here $\partial V \cap \Omega \subset \Gamma \cup (\partial P_0 \cap \Omega)$. The two contributions on ∂P_0—the first term in (6.9d) and the part of the integral over $\partial V \cap \Omega$—are supported on disjoint subsets of ∂P_0. Since $\varphi \le C$ on Γ and $\varphi \le 1$ on Ω, (6.9d) gives $\varphi(y) \le C + \omega_V^y(\partial P_0) \le C + \operatorname{dist}(y, \Gamma_P)/d$, where the last inequality is Lemma 6.5 with $J = \operatorname{int} S$ and $K = \overline{P_0}$.

Fix $z \in \Gamma$. Let D be the component of $\widehat{B}_{P,z} \cap \Omega$ containing z, equivalently, the component of $\widehat{B}_{P,z} \setminus E$ containing z, and let e be an inward unit normal to a face of S containing z. The supporting hyperplane of that face gives $\operatorname{dist}(y, \Gamma_P) \le \langle y - z, e \rangle$ for $y \in \operatorname{int} S$. The preceding estimates therefore give $\varphi(y) \le C + \langle y - z, e \rangle_+ / d$ on $\partial D \cap \Omega$, except possibly at the point of tangency with $\overline{P_0}$, which has ω_D^z-measure zero.

Let $\nu = \omega_D^z|_{\partial D \cap \Omega}$ and let $\sigma_z = \omega^z_{\widehat{B}_{P,z}}$, the normalized surface measure on $\partial \widehat{B}_{P,z}$. Since $\partial D \cap \Omega \subset \partial \widehat{B}_{P,z}$, the first inequality in (6.9a), applied to $D \subset \widehat{B}_{P,z}$, gives $\nu \le \sigma_z$. By (6.8), $\omega_D^z(E \cap \widehat{B}_{P,z}) \ge \eta$, and therefore $\nu(\partial D \cap \Omega) \le 1 - \eta$. Since $\widehat{B}_{P,z} \subset R_P$, we also have $\omega_D^z((E \cap R_P) \cap \partial D) \ge \eta$, while $\omega_D^z((E \cap P_0) \cap \partial D) = 0$. Applying (6.9c) to $D \subset \Omega$ now gives

$$\begin{aligned}\varphi(z) &\le -\theta\eta + \int \varphi \, d\nu \le -\theta\eta + C(1-\eta) + \frac{1}{d}\int \langle y - z, e\rangle_+ \, d\sigma_z(y) \\ &= -\theta\eta + C(1-\eta) + \frac{r}{d}\frac{\omega_{n-1}}{n\omega_n} = (1-\eta)C.\end{aligned}$$

Here $\int \langle y - z, e\rangle_+ \, d\sigma_z(y) = r\omega_{n-1}/(n\omega_n)$, and the last equality uses the identity for $\theta\eta$ above. Taking the supremum over $z \in \Gamma$ and then the maximum with 0 gives $C \le (1-\eta)C$. Since $\eta > 0$, we have $C = 0$, as required. □

Lemma 6.7. *Suppose that P satisfies **Alternative II**. If $x_0 \in \Omega \setminus P$ and $\omega \equiv \omega_\Omega^{x_0}$, then*

$$\omega(P_0) \le W\,\omega(P), \qquad W = \left(1 + \frac{n\,\omega_n}{\omega_{n-1}}\left(\frac{4L}{3M} - 1\right)\eta\right)^{-1}. \tag{6.12}$$

Remark. In [Bou87] and [BG24], this estimate appeared as $\omega(P_0) \le \eta^{-1}(1-\eta)^k\omega(P)$, where k is the number of outer rings. Here the large factor η^{-1} is eliminated, and $W < 1$ decreases as η increases and as Γ_P moves toward ∂P. This allows us to prove $\sup\{\omega(P_0)/\omega(P) : P \text{ satisfies } \textbf{Alternative II}\} < 1$ with $k = 1$. Using one ring instead of several also allows a larger choice of ρ in (6.4).

Proof. By Lemma 6.6, $\omega(P) = \omega(R_P) + \omega(P_0) \ge (1 + \theta^{-1})\omega(P_0)$. Therefore, $\omega(P_0) \le (1 + \theta^{-1})^{-1}\omega(P) = W\omega(P)$. □

Lemma 6.8. *Suppose P satisfies **Alternative II**. Let V and W be defined as in* (6.3)*, so that $V = (\operatorname{vol} P_0)/(\operatorname{vol} P)$. If $W < V$, then for every exponent $\alpha \in (0,1)$, the quantity $\gamma_\alpha := W^\alpha V^{1-\alpha} + (1-W)^\alpha(1-V)^{1-\alpha}$ satisfies $\gamma_\alpha < 1$ and*

$$\sum_{i=0}^{M^n - (M-2)^n} \omega(P_i)^\alpha (\operatorname{vol} P_i)^{1-\alpha} \le \gamma_\alpha\, \omega(P)^\alpha (\operatorname{vol} P)^{1-\alpha}. \tag{6.13}$$

Proof. We may assume that $\omega(P) > 0$, since (6.13) is trivial otherwise. Fix $\alpha \in (0,1)$ and set $H(x) = x^\alpha V^{1-\alpha} + (1-x)^\alpha(1-V)^{1-\alpha}$ for $0 \le x \le 1$. We have $H(V) = 1$ and $H'(x) > 0$ for $0 < x < V$, so H is strictly increasing on $[0, V]$. Put $N = M^n - (M-2)^n$. By Hölder's inequality,

$$\begin{aligned}\sum_{i=0}^{N} \omega(P_i)^\alpha (\operatorname{vol} P_i)^{1-\alpha} &\le \omega(P_0)^\alpha (\operatorname{vol} P_0)^{1-\alpha} + \left(\sum_{i=1}^{N} \omega(P_i)\right)^\alpha \left(\sum_{i=1}^{N} \operatorname{vol} P_i\right)^{1-\alpha} \\ &= H\left(\frac{\omega(P_0)}{\omega(P)}\right)\omega(P)^\alpha (\operatorname{vol} P)^{1-\alpha}.\end{aligned}$$

By Lemma 6.7, $\omega(P_0)/\omega(P) \le W < V$. Therefore, $H(\omega(P_0)/\omega(P)) \le H(W) = \gamma_\alpha < H(V) = 1$, which proves (6.13). □

6.4. **Proof of Theorem 6.1.** Let $n \geq 3$ and $0 < a < 1$. Define η by (6.2). Fix integers $M \geq 3$, $L \geq 2$, and $h \geq n^3$ with $2L \geq 3M$ and $W < V$, where W and V are defined in (6.3). Define ρ_0 by (6.1), and fix $\rho \in (0, \rho_0]$ satisfying (6.4). For $\alpha \in (0,1)$, define γ_α as in Lemma 6.8, and let λ and τ be given as in (6.5) and (6.6).

Let $\Omega \subset \mathbb{R}^n$ be a bounded domain. By translation and dilation, we may assume that $0 \in \Omega$ and $\operatorname{dist}(0, \partial\Omega) > \sqrt{n}$. Thus, every cube of side length at most 1 that meets $\partial\Omega$ lies away from the pole 0. Set $\omega = \omega_\Omega^0$, $E = \partial\Omega$, and $\beta = 1 - 1/L$. Then E is closed, $\omega(\mathbb{R}^n \setminus E) = 0$, and $\beta < 1$. Fix a cube P of side length at most 1.

Suppose first that P satisfies **Alternative I**. By Lemma 6.2 and (6.4), there is a pairwise disjoint cover $\mathcal{Q}$ of $E \cap P$ by cubes Q with $\operatorname{side} Q \leq \beta \operatorname{side} P$ and $\sum_{Q\in\mathcal{Q}} (\operatorname{side} Q)^{n-\rho} \leq (\operatorname{side} P)^{n-\rho}$. Therefore, P satisfies condition (I) of Theorem 4.1.

Suppose instead that P satisfies **Alternative II**. If $\omega(P) = 0$, then condition (II) of Theorem 4.1 holds trivially. Assume that $\omega(P) > 0$. Since ω is supported on E, the cube P meets $\partial\Omega$, so the pole 0 lies outside P by our normalization. Let P_0 be the inner child and $P_1, \ldots, P_N$, $N = M^n - (M-2)^n$, be the outer children. Then $\mathcal{R} = \{P_0, \ldots, P_N\}$ is a pairwise disjoint cover of $E \cap P$. Every cube in $\mathcal{R}$ has side length at most $\operatorname{side} P_0 = (1 - 2/M) \operatorname{side} P \leq \beta \operatorname{side} P$, while its $(n-\rho)$-dimensional cubical content is $\tau(\operatorname{side} P)^{n-\rho}$. Thus, $\tau > 1$ by Principle 3.3. Finally, Lemma 6.8 gives $\sum_{R\in\mathcal{R}} \omega(R)^\alpha (\operatorname{vol} R)^{1-\alpha} \leq \gamma_\alpha \omega(P)^\alpha (\operatorname{vol} P)^{1-\alpha}$. Therefore, P satisfies condition (II) of Theorem 4.1 with exponent α.

Fix $\alpha \in (0,1)$. We have shown that every cube of side length at most 1 satisfies condition (I) or (II) of Theorem 4.1. Hence $\overline{\dim}_H \omega \leq n - \lambda_\alpha \rho/(1+\lambda_\alpha)$, where $\lambda_\alpha = -\log_\tau(\gamma_\alpha)/(1-\alpha)$. It remains to let $\alpha \to 1$. Put $f(\alpha) = \log \gamma_\alpha$. The function f is convex on $[0,1]$, since it is the logarithm of a sum of exponentials of affine functions of α, and $f(1) = 0$. Therefore, the slope of the secant line joining $(\alpha, f(\alpha))$ to $(1, f(1))$, namely $[f(1) - f(\alpha)]/(1-\alpha)$, is nondecreasing as $\alpha \to 1$ and converges to $f'(1)$. Direct differentiation gives $f'(1) = W \log(W/V) + (1-W)\log((1-W)/(1-V))$. Thus, λ_α increases to λ defined in (6.5). Since $x \mapsto x/(1+x)$ is continuous and increasing, letting $\alpha \to 1$ yields $\overline{\dim}_H \omega \leq n - \lambda\rho/(1+\lambda)$. Taking the supremum over bounded domains gives $d_n \leq n - \lambda\rho/(1+\lambda)$. This concludes the proof of Theorem 6.1.

7. Lower bounds on Bourgain's constant b_n

We now derive explicit lower bounds for Bourgain's constant $b_n = n - d_n$ from Theorem 6.1. We first treat $n \geq 6$ uniformly and then bound b_3, b_4, and b_5 directly.

Theorem 7.1. *For all $n \geq 6$,*

$$b_n \geq \left(\frac{1}{n-1}\right)^{2n+6.12\sqrt{n}}. \tag{7.1}$$

Proof. Fix $n \geq 6$. Set $a = n/(n+1)$ and $h = n^6$, define η by (6.2), and put $q := \omega_{n-1}/(\omega_n \eta)$, $m := n + \lceil q \rceil + 1$, $M := 2m$, and $L := 3m$. Then $2L = 3M$, $W = 1 - n/(n+q)$, and $V = (1 - 1/m)^n$.

Step 1: $W < V$. Lemma 3.7 gives

$$q \le \frac{n^{3/2}}{2\sqrt{2\pi}}\, e^{1/(6n+12)} \left(1+\frac{2}{n}\right)^{1/2} \left(1+\frac{1}{n}\right) \frac{1+n^{-6}}{1-2^{2-n}} < 0.3\, n^{3/2},$$

where the last inequality follows by bounding the n-dependent factors by their values at $n = 6$. Hence $m < n + 0.3n^{3/2} + 2 < n^{3/2}$. Since $m > n + q$, Bernoulli's inequality gives $V \ge 1 - n/m > W$. Moreover, since $m - n - q \ge 1$ and $n + q < m < n^{3/2}$,

$$V - W \ge \frac{n(m-n-q)}{m(n+q)} > \frac{n}{m^2} > \frac{1}{n^2}.$$

Step 2: an admissible ρ. Let $\Sigma(\rho)$ denote the left hand side of (6.4). We estimate $\Sigma(\rho)$ by linearizing at $\rho = 0$. Direct substitution and Bernoulli's inequality give

$$1 - \Sigma(0) = \frac{\omega_n}{(4L)^n}\left[\left(1 - \frac{2\sqrt{n}}{h}\right)^n - a\right] \ge \frac{\omega_n}{2(n+1)(4L)^n}.$$

Indeed, the expression in brackets is at least $1/(n+1) - 2n^{-9/2} \ge 1/[2(n+1)]$.

At $\rho = 0$, the first two terms of Σ sum to $1 - 2^{n-1}/L^n$. Since $\Sigma(0) < 1$, the last three terms sum to less than $2^{n-1}/L^n$. Differentiating (6.4) termwise gives

$$\Sigma'(t) = \underbrace{\log\frac{L}{L-1}\left(1-\frac{1}{L}\right)^{n-t}}_{\text{I}} + \underbrace{\log L\left(L^n - (L-1)^n - 2^{n-1}\right)L^{t-n}}_{\text{II}}$$
$$+ \underbrace{\boxed{\begin{aligned}&\log(2L)\,2^{n-1}(2^n-1)(2L)^{t-n}\\ &+\log(2Lh)\left(2^{n-1}h^n - \frac{\omega_n}{2^n}(h-2\sqrt{n})^n\right)(2Lh)^{t-n} + a\omega_n \log(4L)(4L)^{t-n}\end{aligned}}}_{\text{III}}.$$

For $0 \le t \le \rho_0$, we have $\text{I} \le 2/L$, while Bernoulli's inequality gives $\text{II} \le (2Lh)^t (n/L)\log L$. At $t = 0$, the three terms comprising III have total coefficient less than $2^{n-1}/L^n$; since their bases are at most $2Lh$, it follows that

$$\text{III} \le (2Lh)^t \frac{2^{n-1}}{L^n}\log(2Lh).$$

Therefore,

$$\Sigma'(t) \le (2Lh)^t\left[\frac{2}{L} + \frac{n}{L}\log L + \frac{2^{n-1}}{L^n}\log(2Lh)\right] < \frac{n}{2}.$$

Set

$$\rho := \frac{\omega_n}{2n^2(4L)^n}. \tag{7.2}$$

Since $\omega_n \le 2^n$ and $L \ge 3n$, we have $\rho \le 1/[2n^2(6n)^n] < 1/(30n^6 \log n) = \rho_0$. By the mean value theorem,

$$\Sigma(\rho) - \Sigma(0) < \frac{n}{2}\rho = \frac{\omega_n}{4n(4L)^n} \le \frac{\omega_n}{2(n+1)(4L)^n} \le 1 - \Sigma(0).$$

Thus, (6.4) holds.

Step 3: a lower bound for λ. Put $\Phi(x) := x\log(x/V) + (1-x)\log((1-x)/(1-V))$. Then $\Phi(V) = \Phi'(V) = 0$ and $\Phi''(x) = 1/[x(1-x)] \geq 4$, so Taylor's theorem and Step 1 give $\Phi(W) \geq 2(W-V)^2 > 1/n^4$. Also, $\tau \leq (2m)^\rho$, and since $2m < n^2$ and $\rho \leq \rho_0$, we have $\log\tau \leq \rho_0 \log(n^2) = 1/(15n^6) < 1/n^4$. By (6.5), $\lambda > 1$, and hence $\lambda/(1+\lambda) > 1/2$.

Step 4: comparison with (7.1). By Theorem 6.1 and (7.2), $b_n > \rho/2 = \omega_n/[4n^2(4L)^n]$. Lemma 3.7 gives

$$\omega_n \geq \frac{e^{1-1/(6n+12)}}{\sqrt{\pi(n+2)}}\left(\frac{\sqrt{2\pi e}}{\sqrt{n+2}}\right)^n > \frac{1}{n^2}\left(\frac{4}{\sqrt{n+2}}\right)^n.$$

Here we used $\sqrt{2\pi e} > 4$ and $n^2 > \sqrt{\pi(n+2)}$. Consequently,

$$b_n > \frac{1}{4n^4}\left(\frac{1}{L\sqrt{n+2}}\right)^n.$$

Since $L = 3m < n^{3/2} + 3n + 6$, put $A := \sqrt{1+2/n}\,(1 + 3/\sqrt{n} + 6n^{-3/2})$. Then $L\sqrt{n+2} < n^2 A$, while $\sqrt{n}\log A \leq 1/\sqrt{n} + 3 + 6/n$. It therefore suffices to show

$$6.12\log(n-1) > \frac{2\sqrt{n}}{n-1} + \frac{1}{\sqrt{n}} + 3 + \frac{6}{n} + \frac{4\log n + \log 4}{\sqrt{n}}.$$

For $n \geq 6$, the left hand side is greater than $6.12\log 5 > 9.8$. Using $\log n/\sqrt{n} \leq 2/e$, the right hand side is at most $2\sqrt{6}/5 + 1/\sqrt{6} + 4 + 8/e + \log 4/\sqrt{6} < 9$. This proves (7.1). □

Theorem 7.2. *We have*

$$b_3 \geq 1.0118 \times 10^{-5}, \qquad b_4 \geq 1.206 \times 10^{-7}, \qquad b_5 \geq 8.006 \times 10^{-10}.$$

In particular, the upper Hausdorff dimension of harmonic measure on any domain in $\mathbb{R}^3$ *is less than* 2.99999.

Proof. Find a calculator and apply Theorem 6.1 with

n	M	L	a	h	ρ
3	9	17	117/200	10^5	1.05628×10^{-5}
4	10	17	13/20	10^5	1.22747×10^{-7}
5	11	19	143/200	10^6	8.0084×10^{-10}.

In each row $2L > 3M$. Let $\Sigma(\rho)$ denote the left hand side of (6.4). By direct evaluation,

n	W	V	ρ_0	$1 - \Sigma(\rho)$	$\lambda\rho/(1+\lambda)$
3	0.4581...	0.4705...	$2.3829... \times 10^{-5}$	$1.7678... \times 10^{-11}$	$1.01183... \times 10^{-5}$
4	0.4073...	0.4096	$4.2411... \times 10^{-6}$	$6.6680... \times 10^{-13}$	$1.206... \times 10^{-7}$
5	0.3650...	0.3666...	$9.2649... \times 10^{-7}$	$7.6165... \times 10^{-15}$	$8.006... \times 10^{-10}$.

Thus, $W < V$, $\rho < \rho_0$, and (6.4) holds in each case. The conclusion follows. □

Proof of Theorem 1.1. When $n = 2$, both sides of (1.5) equal 1 by the Jones–Wolff theorem. For $n = 3, 4, 5$, Theorem 7.2 gives, respectively,

$$b_3 \geq 1.0118 \times 10^{-5} > 1.006\ldots \times 10^{-5} = 2^{-(6+6.12\sqrt{3})},$$
$$b_4 \geq 1.206 \times 10^{-7} > 2.203\ldots \times 10^{-10} = 3^{-(8+6.12\sqrt{4})},$$
$$b_5 \geq 8.006 \times 10^{-10} > 5.500\ldots \times 10^{-15} = 4^{-(10+6.12\sqrt{5})}.$$

For $n \geq 6$, (1.5) is Theorem 7.1. □

Proposition 7.3 (optimality of the leading exponent)**.** *Let B_n denote the supremum of the lower bounds on b_n furnished by Theorem 6.1 over all admissible choices of its parameters. Then*

$$B_n \leq \left(\frac{1}{n-1}\right)^{2n-o(n)} \qquad \text{as } n \to \infty.$$

Thus, the coefficient 2 in the leading term of the exponent in Theorem 1.1 cannot be improved by a different choice of parameters in Theorem 6.1.

Proof. Since $\lambda\rho/(1+\lambda) < \rho$, it suffices to bound every admissible ρ from above. We first show that $W < V$ forces $L \gtrsim n^{3/2}$. Put $x = 4L/(3M) - 1 \geq 1$. Since $\eta < 2/n$, the inequality $W < V$ and Bernoulli's inequality give

$$1 + 2\frac{\omega_n}{\omega_{n-1}}x > 1 + \frac{n\omega_n}{\omega_{n-1}}x\eta > \left(1 - \frac{2}{M}\right)^{-n} \geq 1 + \frac{2n}{M}.$$

Thus, $Mx > n\omega_{n-1}/\omega_n$, and hence $L = \frac{3}{4}M(1+x) > \frac{3}{4}n\omega_{n-1}/\omega_n \gtrsim n^{3/2}$, where the last estimate follows from Lemma 3.7.

Let $\Sigma(\rho)$ denote the left hand side of (6.4). At $\rho = 0$,

$$1 - \Sigma(0) = \frac{\omega_n}{(4L)^n}\left[\left(1 - \frac{2\sqrt{n}}{h}\right)^n - a\right] < \frac{\omega_n}{(4L)^n}.$$

The second term of $\Sigma(\rho)$ is $c_2 L^\rho$, where

$$c_2 = 1 - \left(1 - \frac{1}{L}\right)^n - \frac{2^{n-1}}{L^n} = \frac{n}{L}(1 + o(1)),$$

uniformly over admissible parameters, since $L \gtrsim n^{3/2}$. All five terms of Σ are nondecreasing in ρ, and $L^\rho - 1 \geq \rho \log L$. Therefore, (6.4) implies

$$\rho\,\frac{n}{L}(1 + o(1))\log L \leq \Sigma(\rho) - \Sigma(0) \leq 1 - \Sigma(0) < \frac{\omega_n}{(4L)^n}.$$

Consequently,

$$\rho \leq \frac{L}{n\log L}(1 + o(1))\frac{\omega_n}{(4L)^n}.$$

Taking nth roots and using $\omega_n^{1/n} = n^{-1/2+o(1)}$ and $L \gtrsim n^{3/2}$ gives

$$\rho^{1/n} \leq \frac{\omega_n^{1/n}}{4}\, L^{-1+1/n}(n\log L)^{-1/n}(1 + o(1)) \leq n^{-2+o(1)}.$$

Thus, every lower bound furnished by Theorem 6.1 is at most $n^{-2n+o(n)} = (n-1)^{-2n+o(n)}$, which proves the proposition. □

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